

BFS Traversal

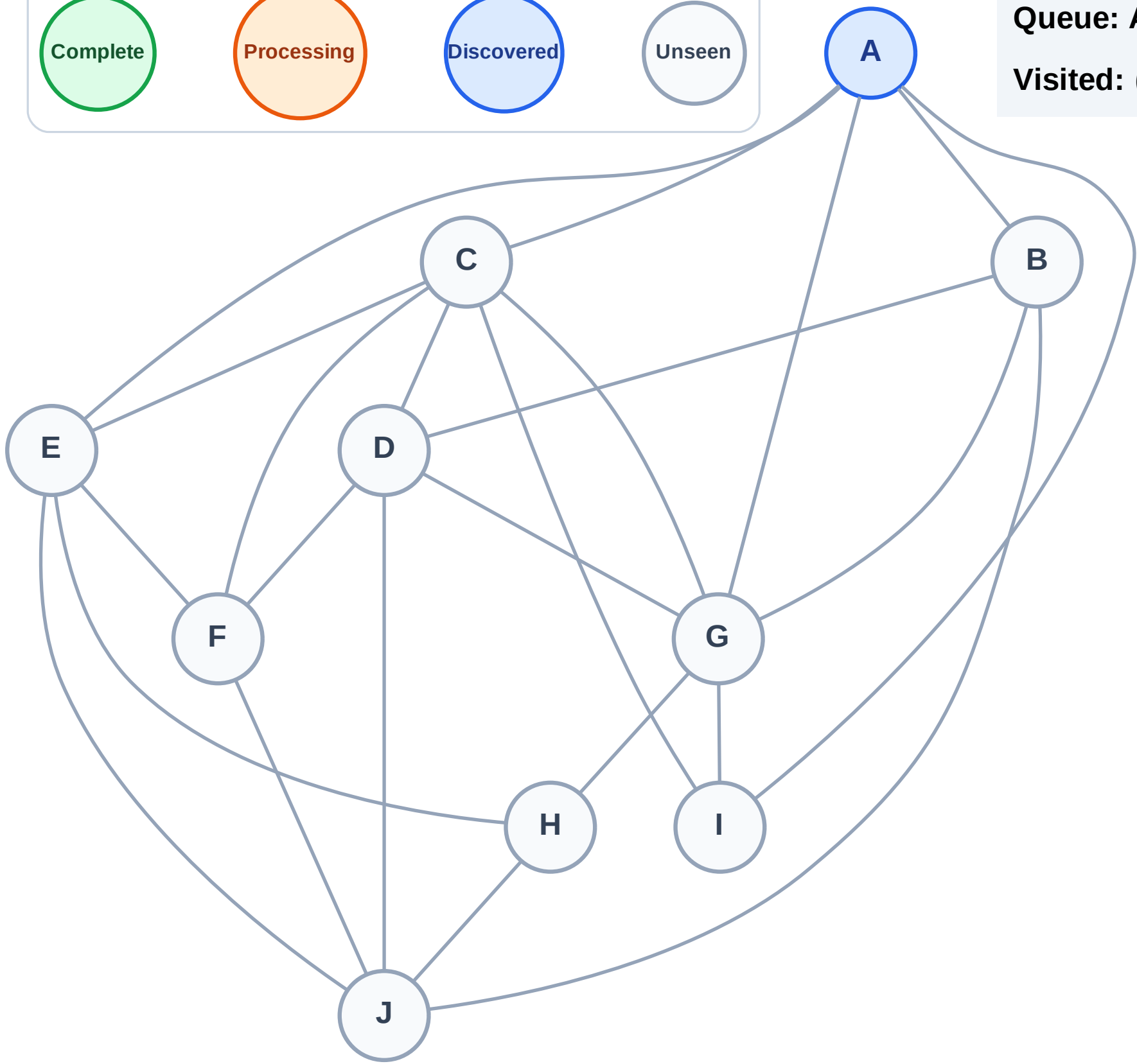
Step 1: Start at A

Legend



Queue: A

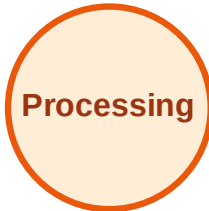
Visited: (none)



BFS Traversal

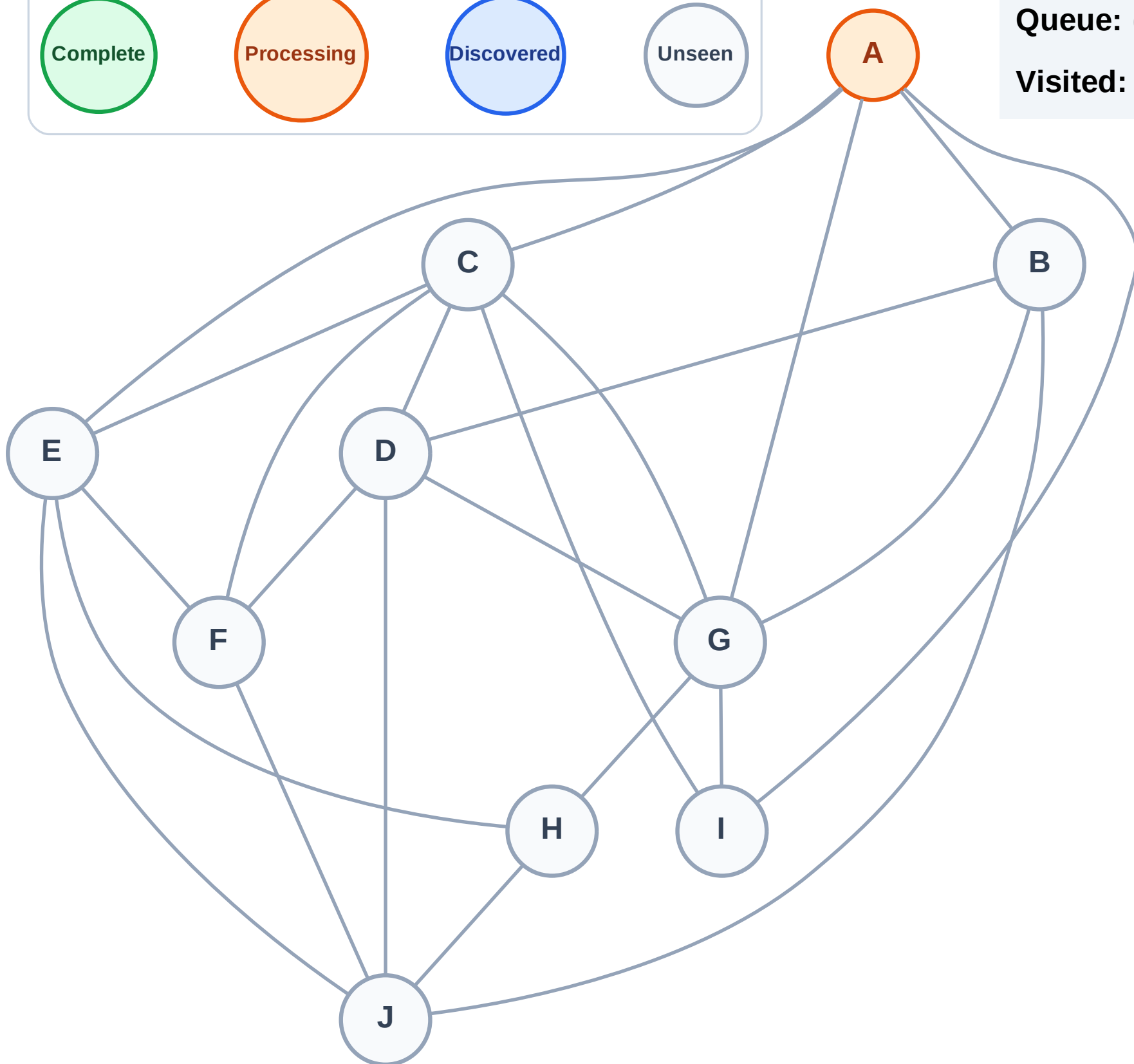
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, C, E, G, I from A

Legend

Complete

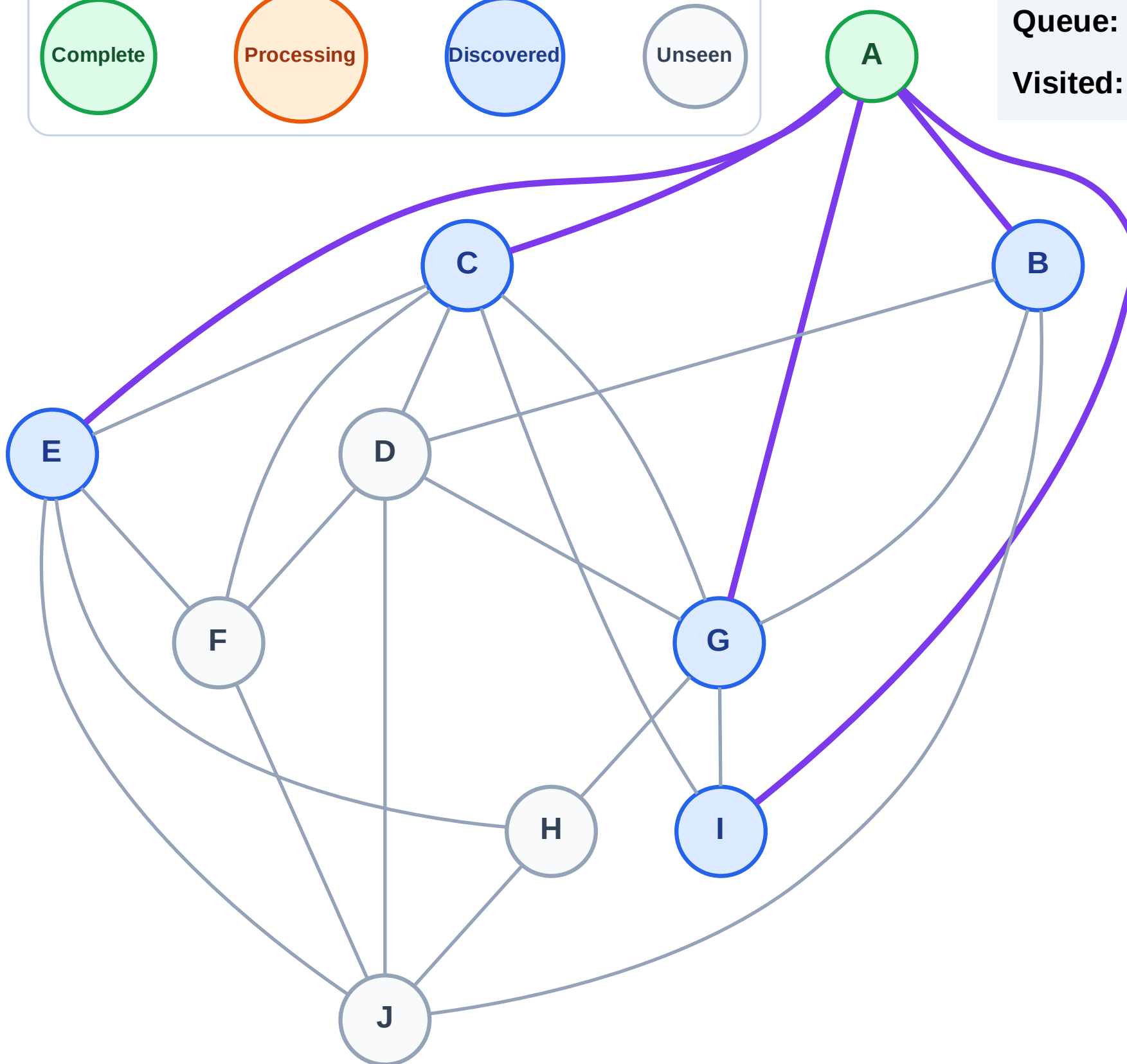
Processing

Discovered

Unseen

Queue: B → C → E → G → I

Visited: A



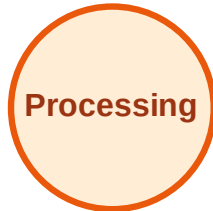
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



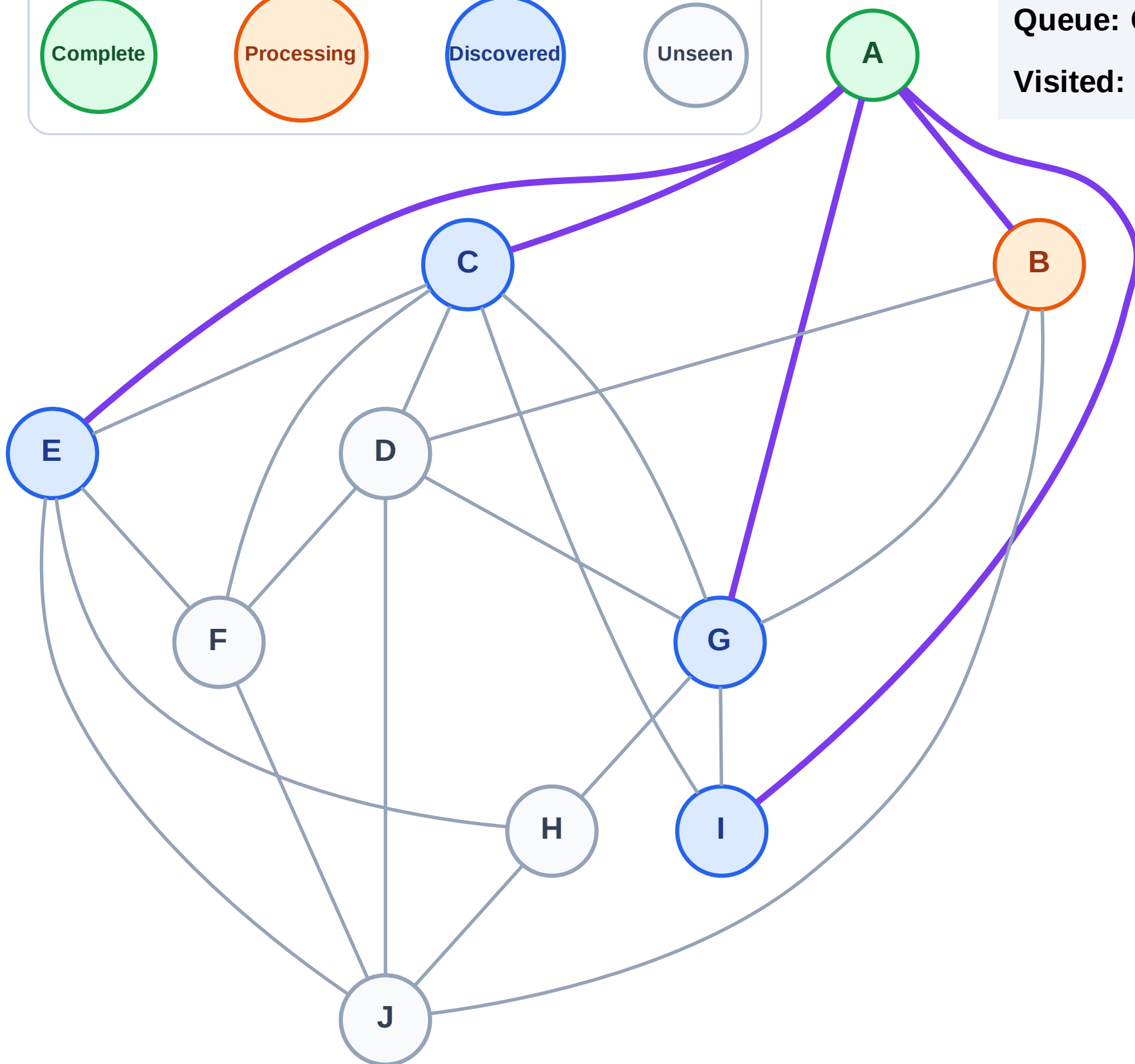
Discovered



Unseen

Queue: C → E → G → I

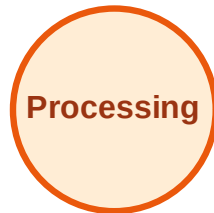
Visited: A



BFS Traversal

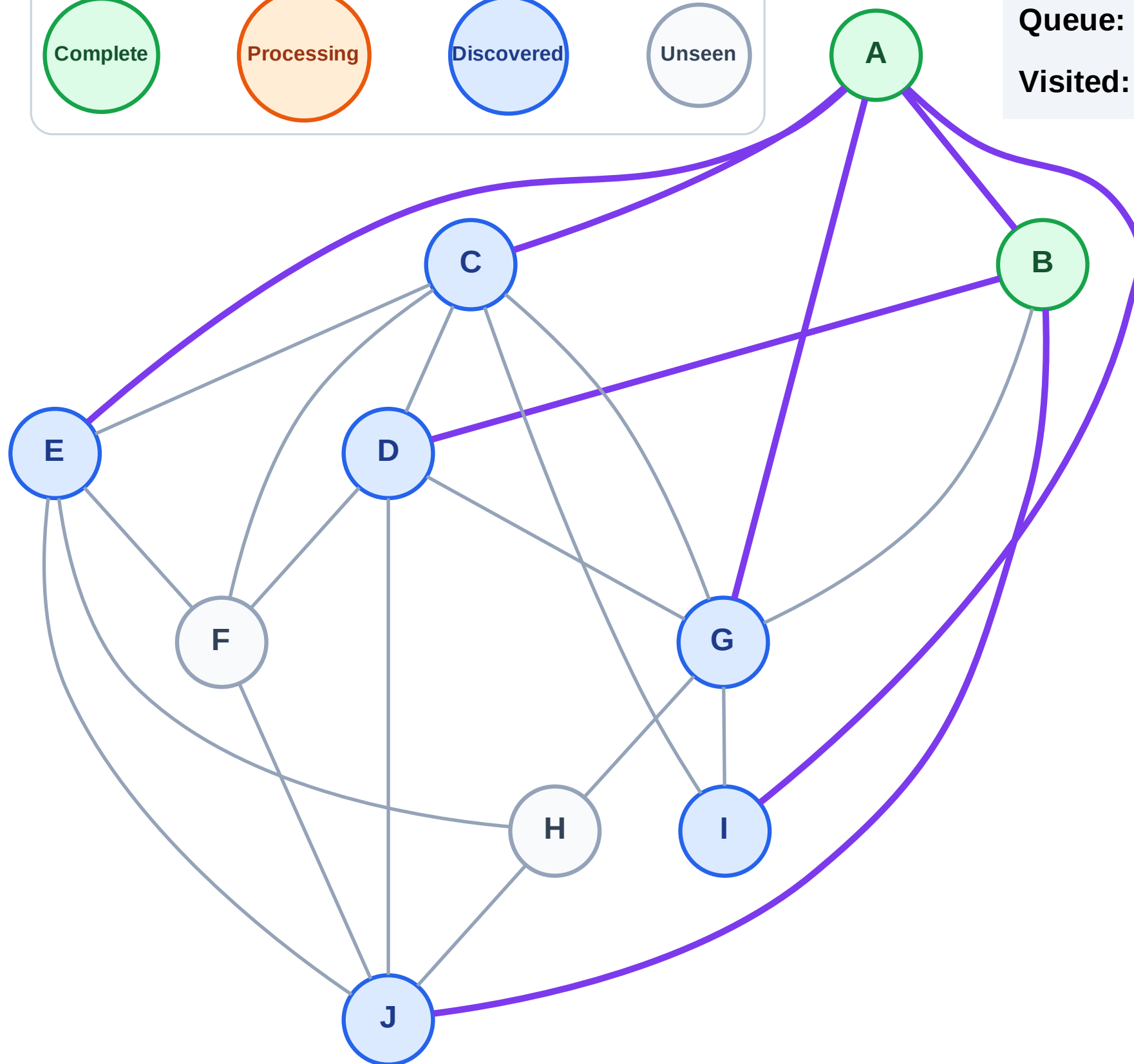
Step 5: Discover D, J from B

Legend



Queue: C → E → G → I → D → J

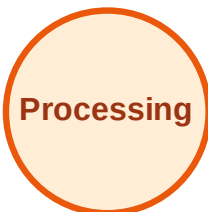
Visited: A, B



BFS Traversal

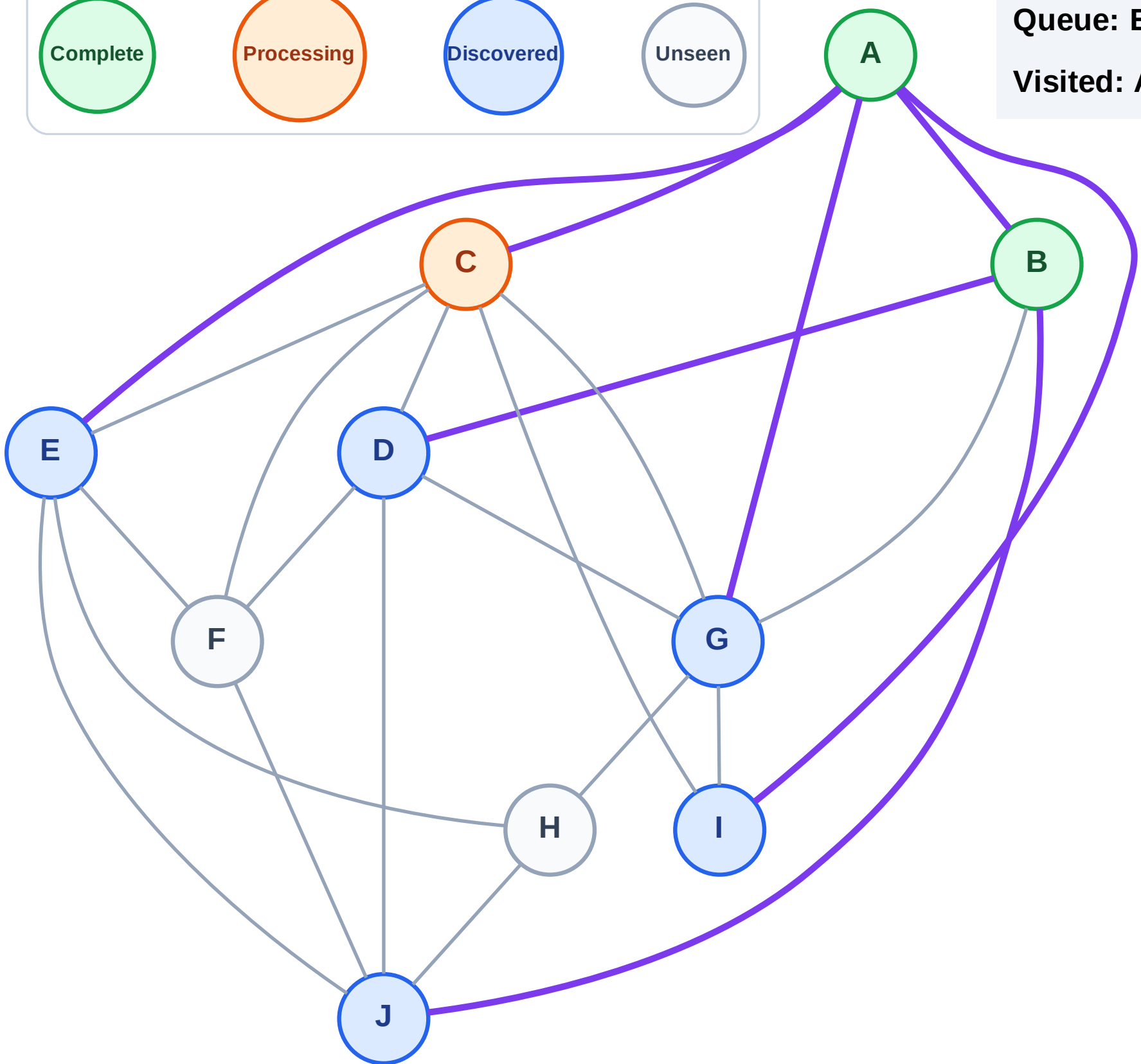
Step 6: Process node C

Legend



Queue: E → G → I → D → J

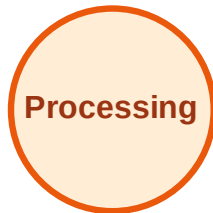
Visited: A, B



BFS Traversal

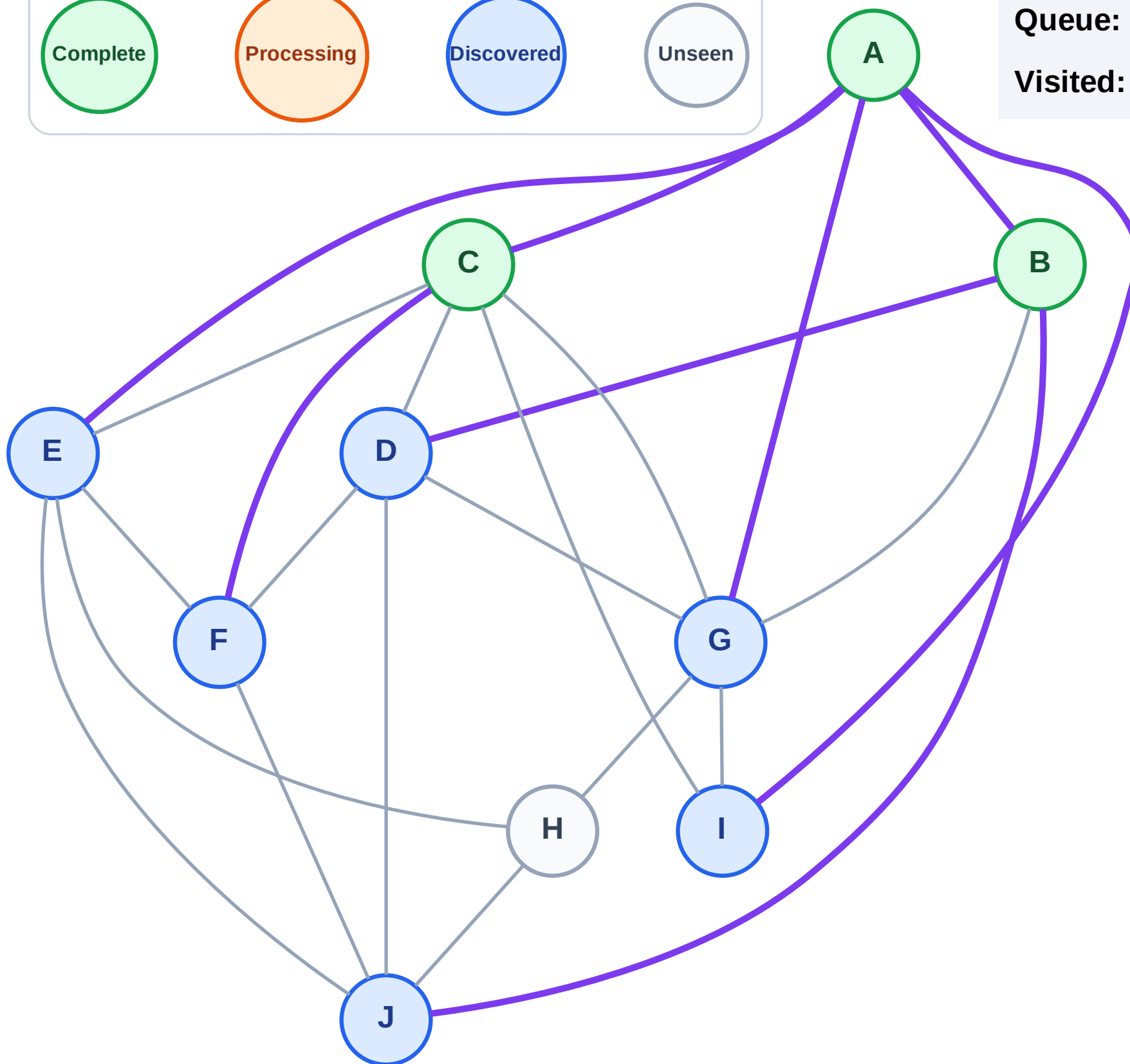
Step 7: Discover F from C

Legend



Queue: E → G → I → D → J → F

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

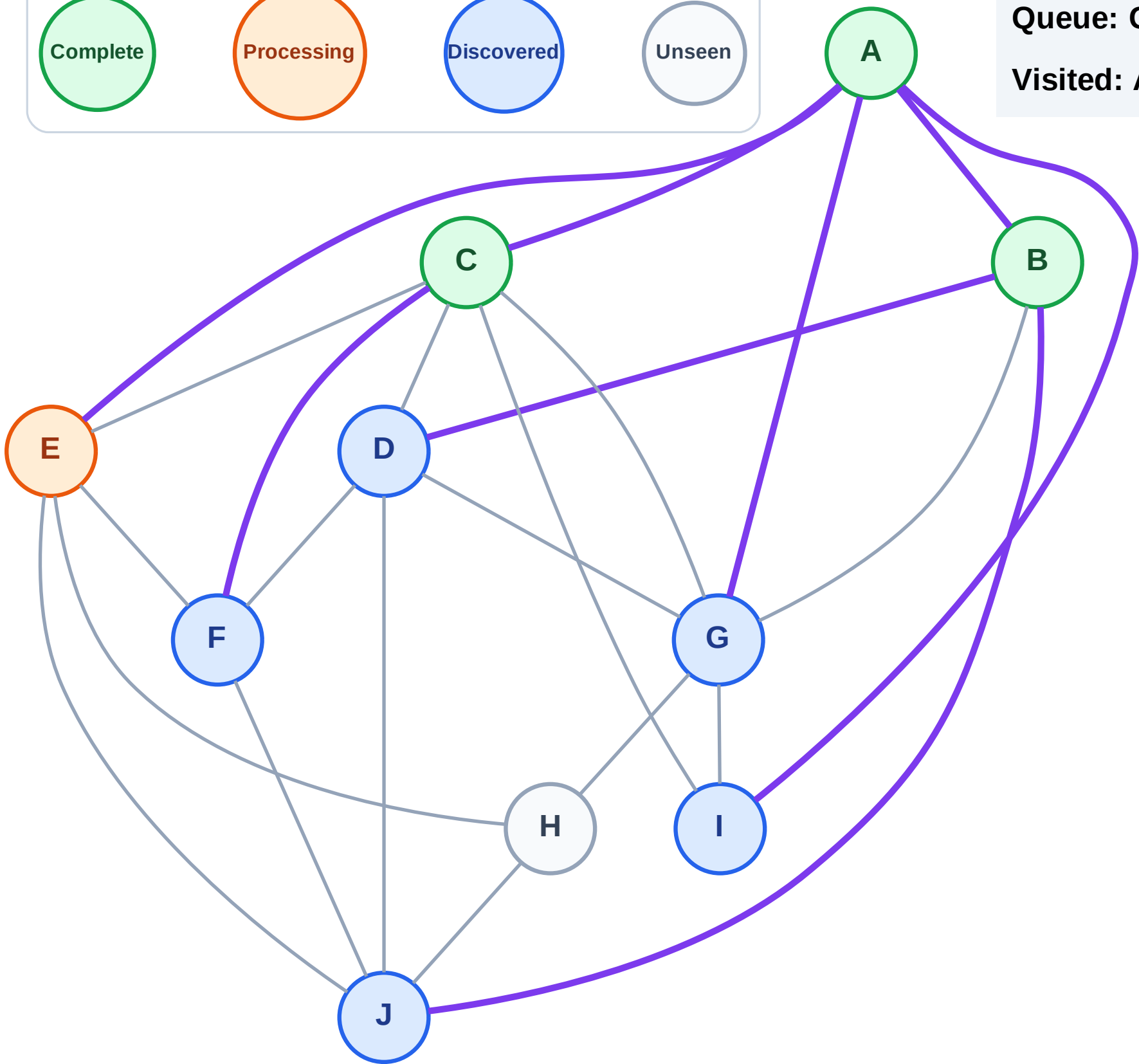
Processing

Discovered

Unseen

Queue: G → I → D → J → F

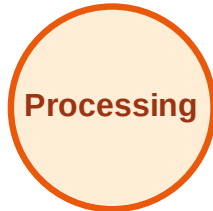
Visited: A, B, C



BFS Traversal

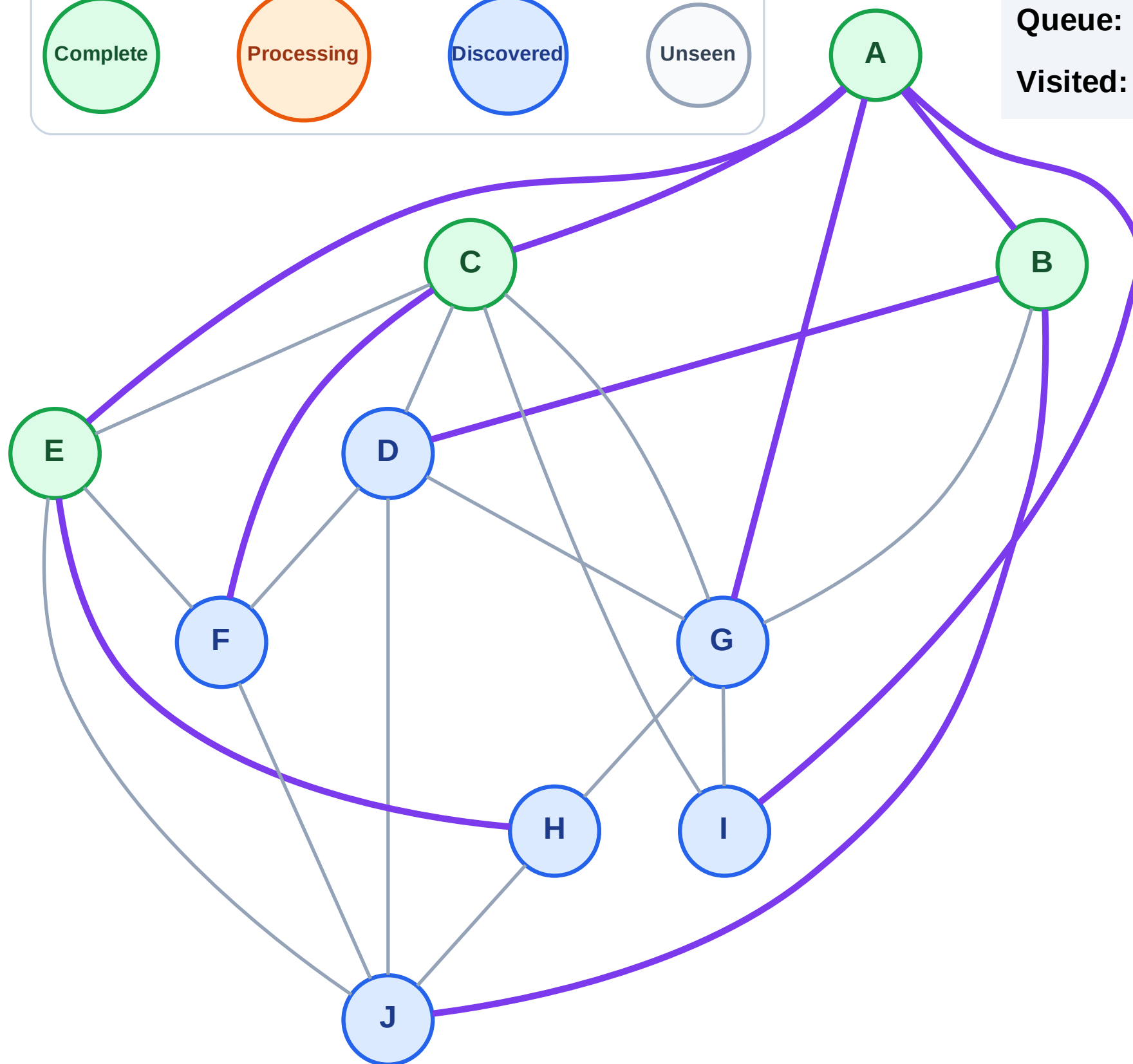
Step 9: Discover H from E

Legend



Queue: G → I → D → J → F → H

Visited: A, B, C, E



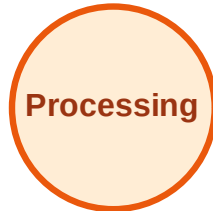
BFS Traversal

Step 10: Process node G

Legend



Complete



Processing



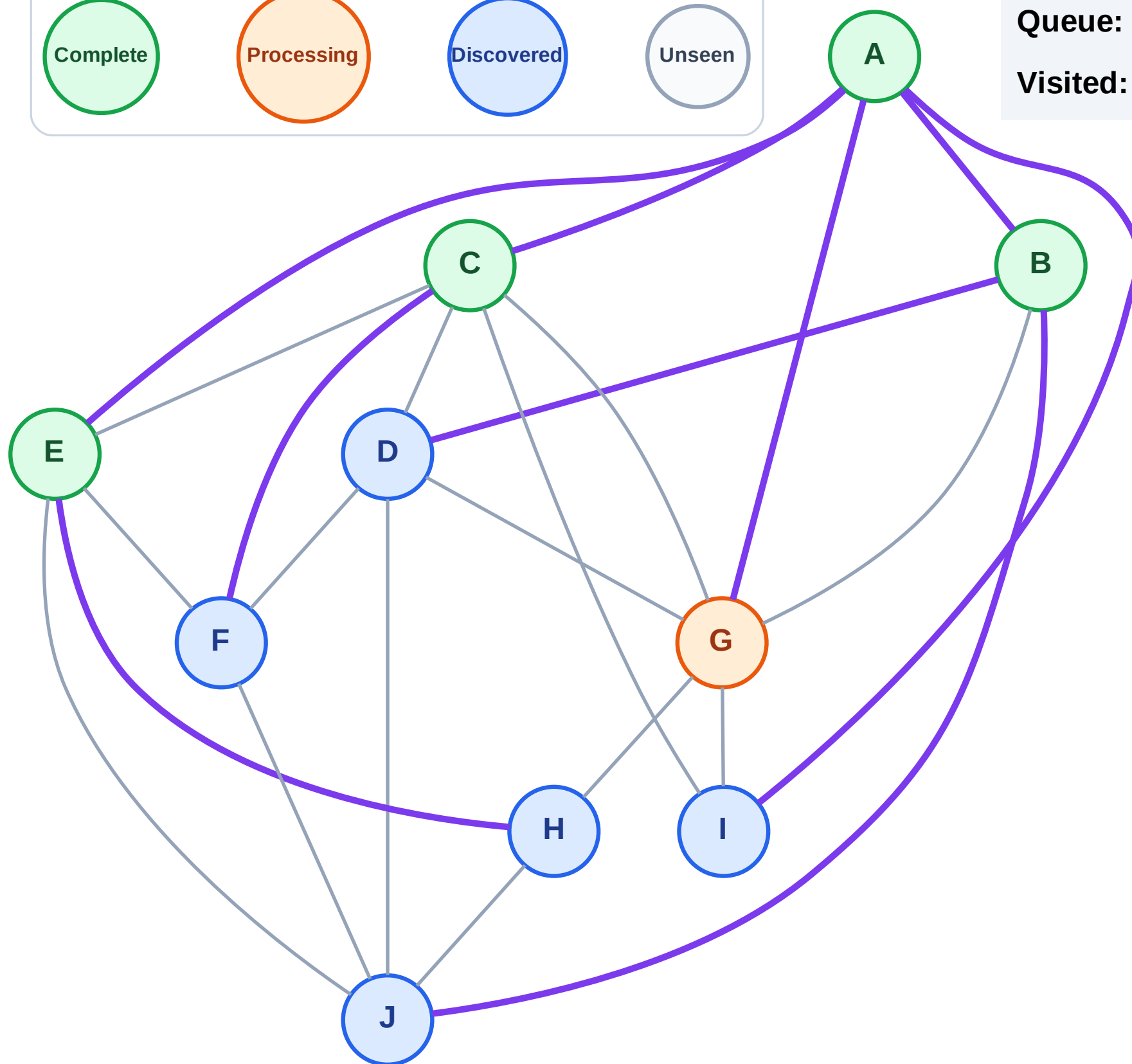
Discovered



Unseen

Queue: I → D → J → F → H

Visited: A, B, C, E



BFS Traversal

Step 11: No new neighbors from G

Legend

Complete

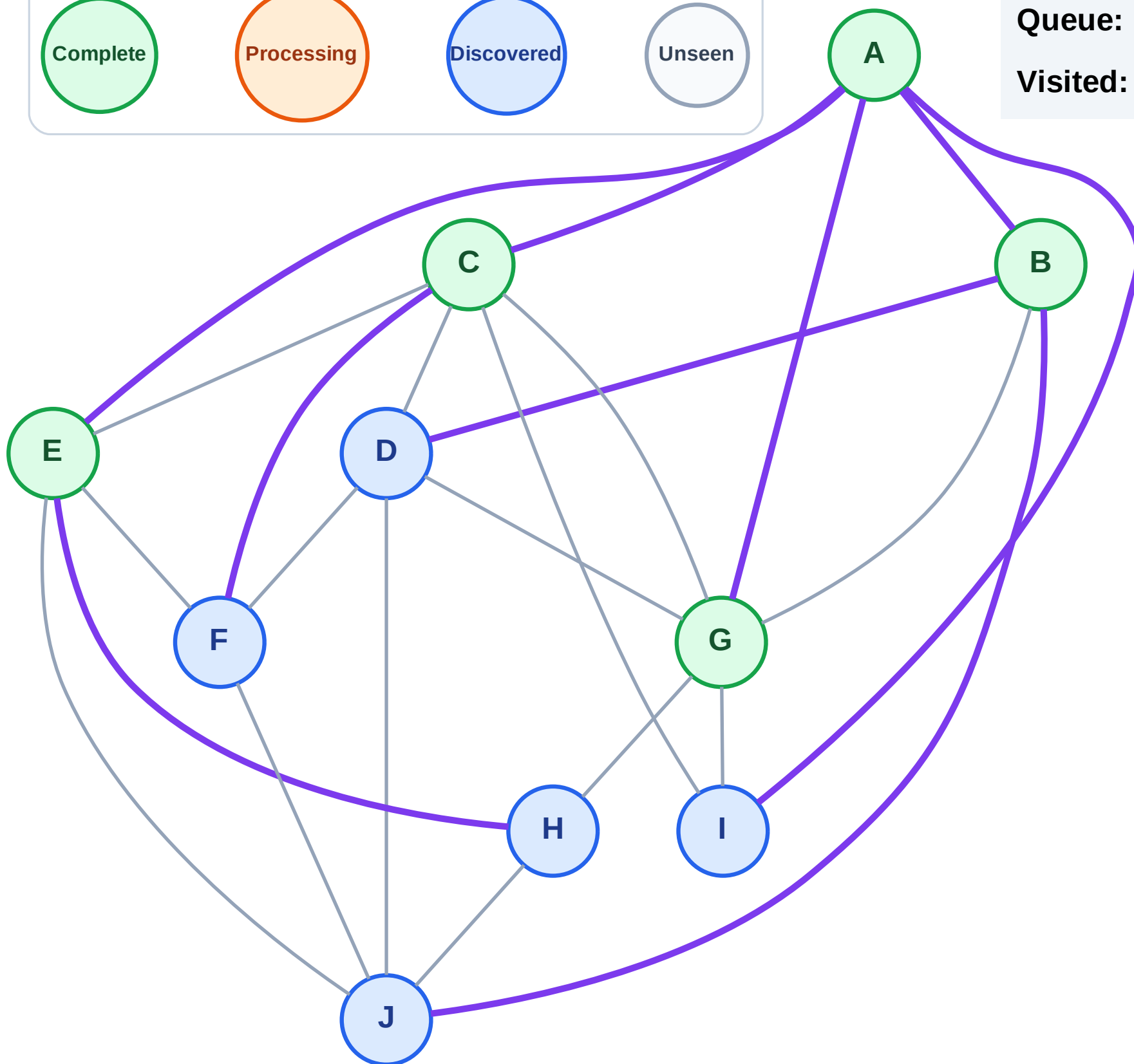
Processing

Discovered

Unseen

Queue: I → D → J → F → H

Visited: A, B, C, E, G



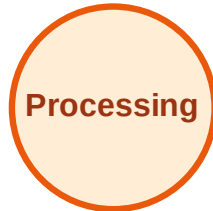
BFS Traversal

Step 12: Process node I

Legend



Complete



Processing



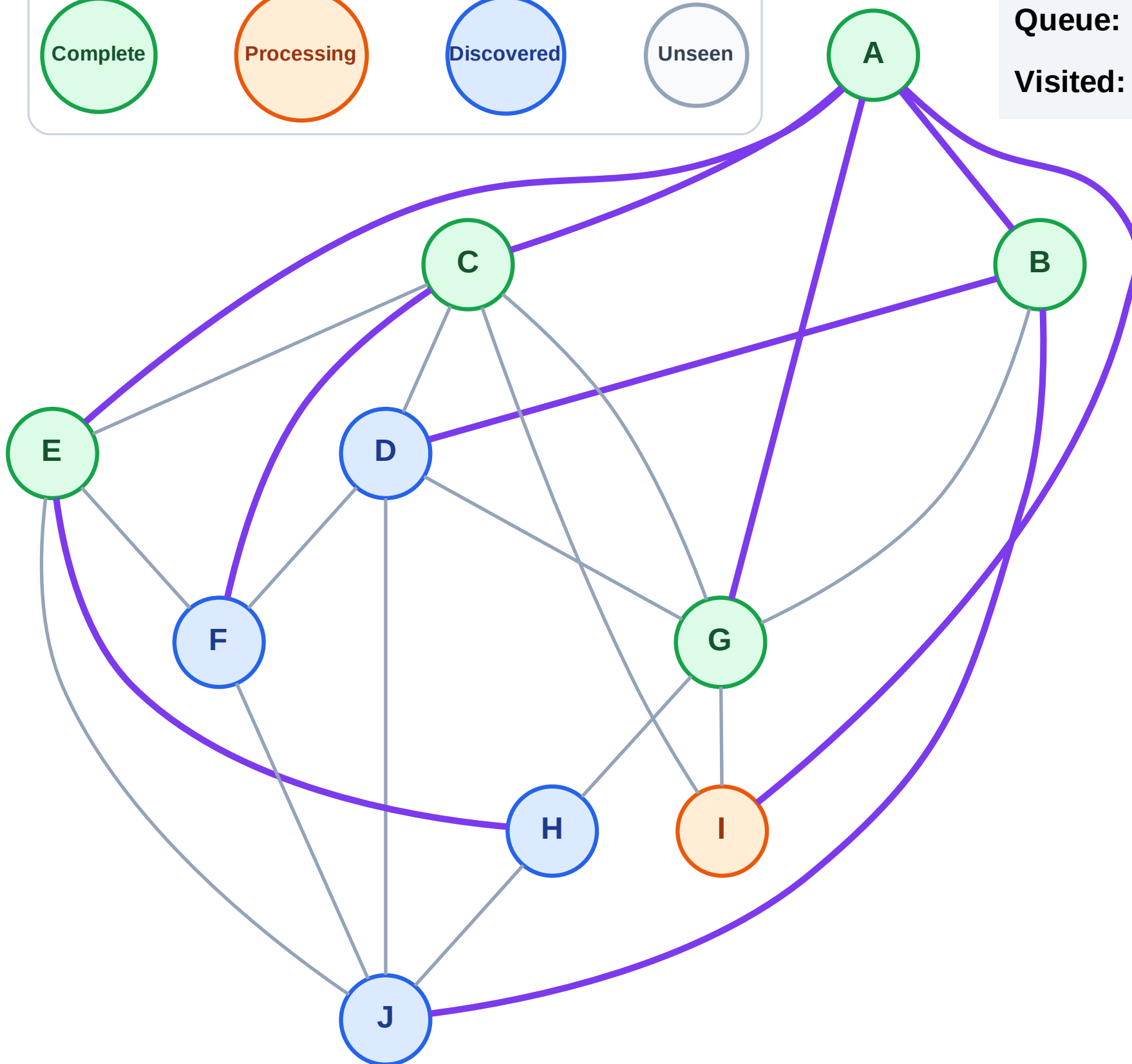
Discovered



Unseen

Queue: D → J → F → H

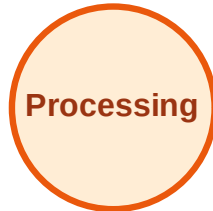
Visited: A, B, C, E, G



BFS Traversal

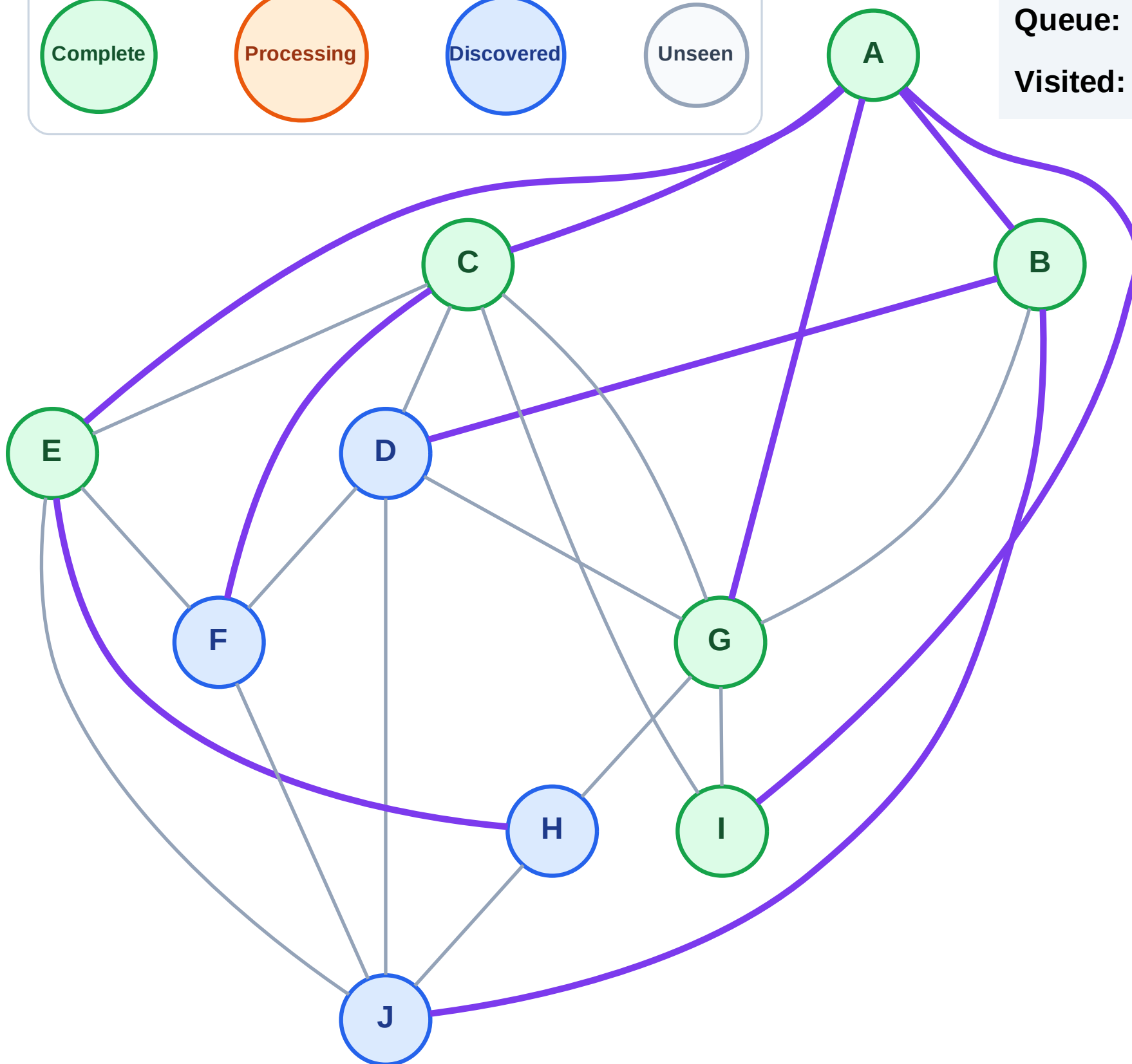
Step 13: No new neighbors from I

Legend



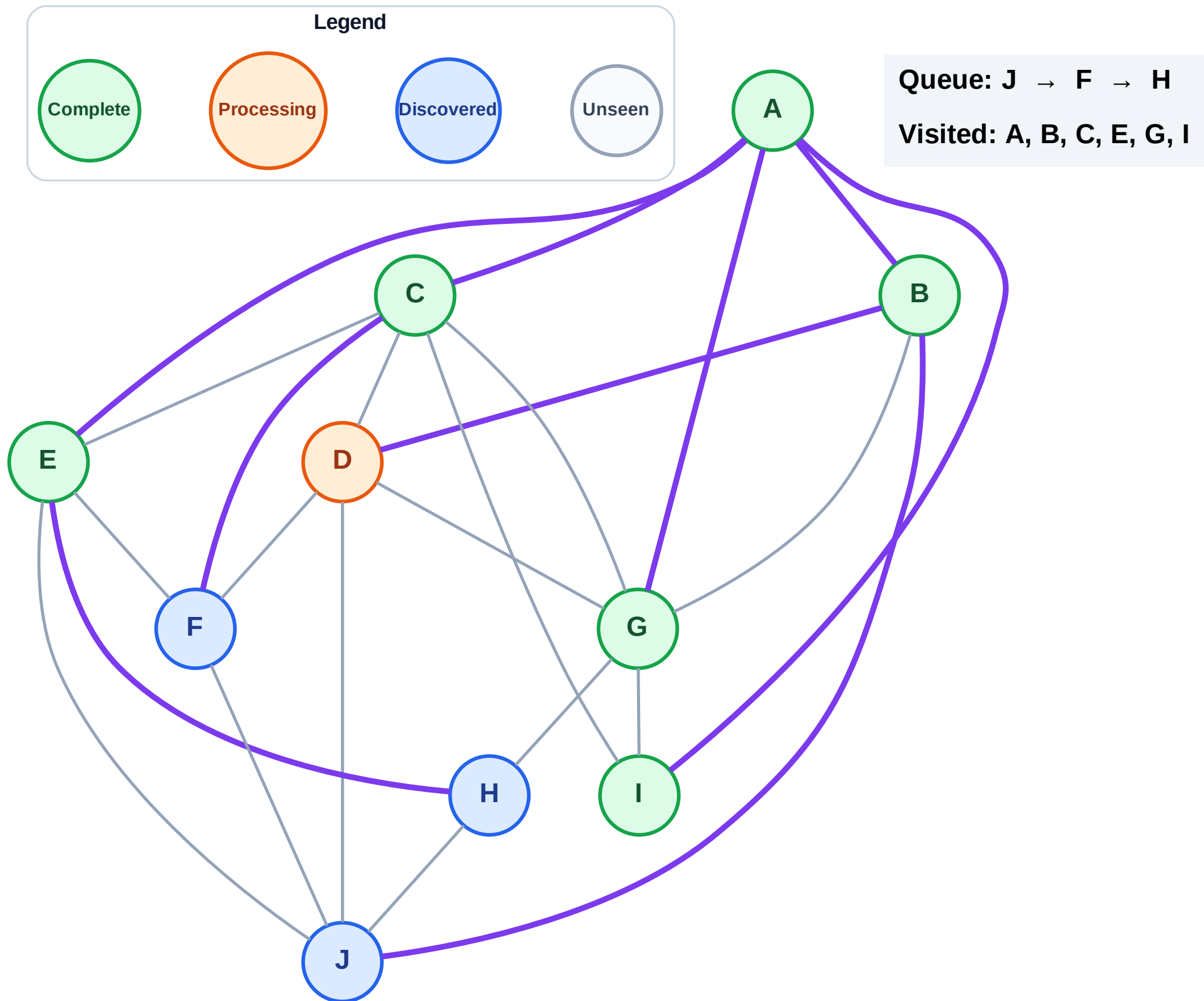
Queue: D → J → F → H

Visited: A, B, C, E, G, I



Step 14: Process node D

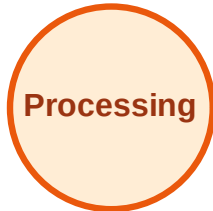
Step 14: Process node D



BFS Traversal

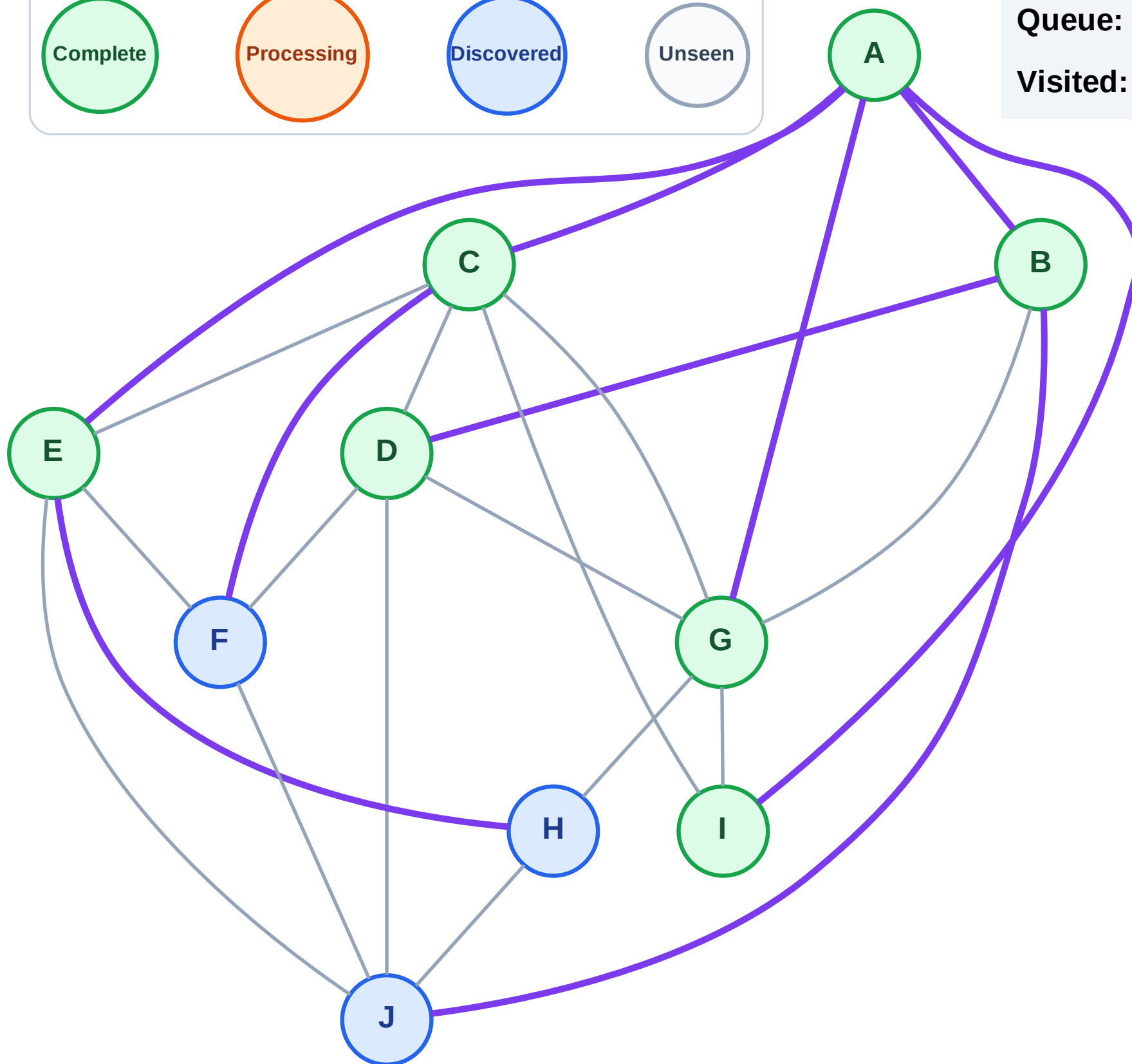
Step 15: No new neighbors from D

Legend



Queue: J → F → H

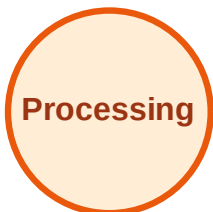
Visited: A, B, C, D, E, G, I



BFS Traversal

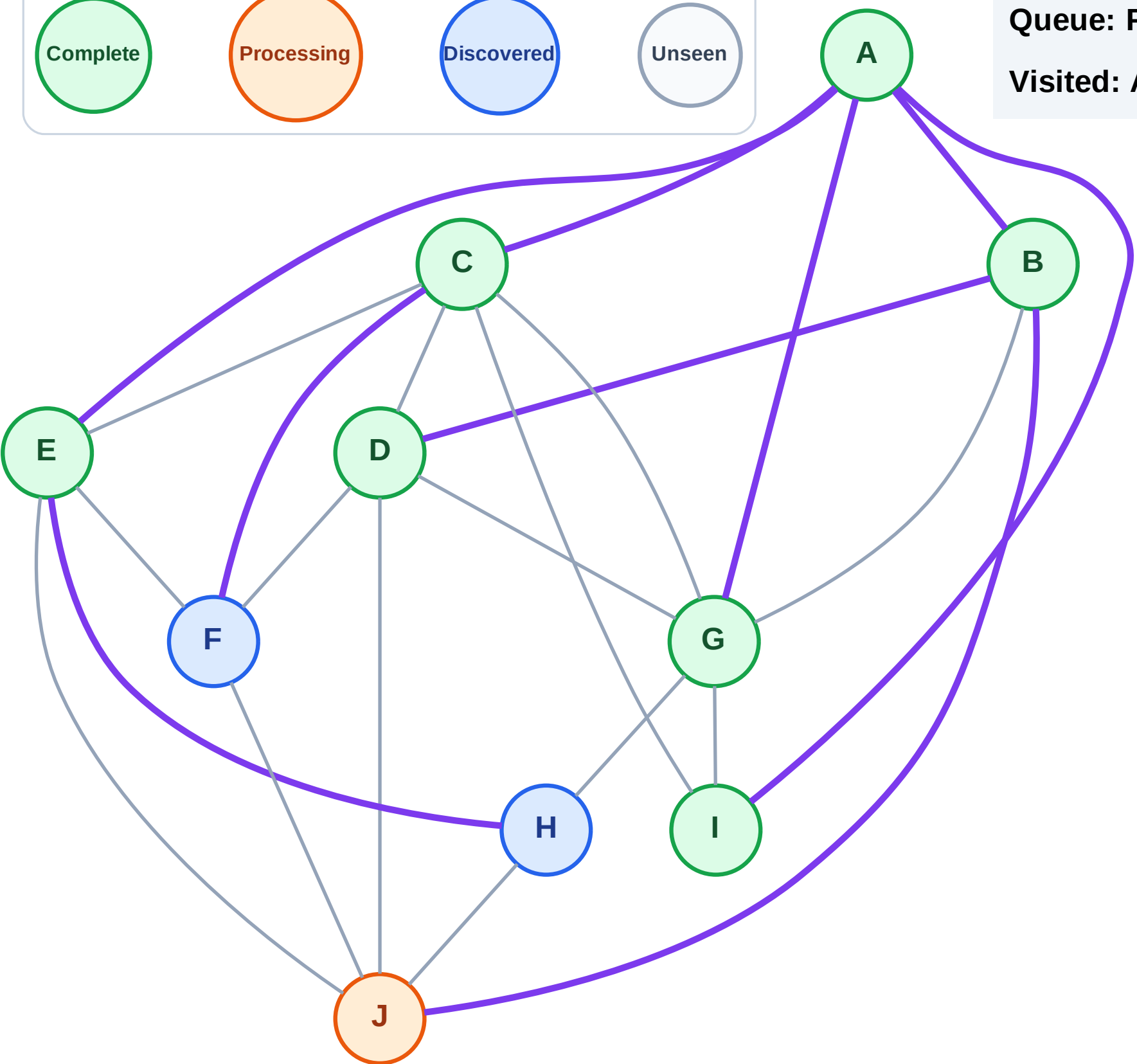
Step 16: Process node J

Legend



Queue: F → H

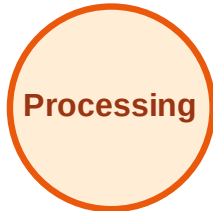
Visited: A, B, C, D, E, G, I



BFS Traversal

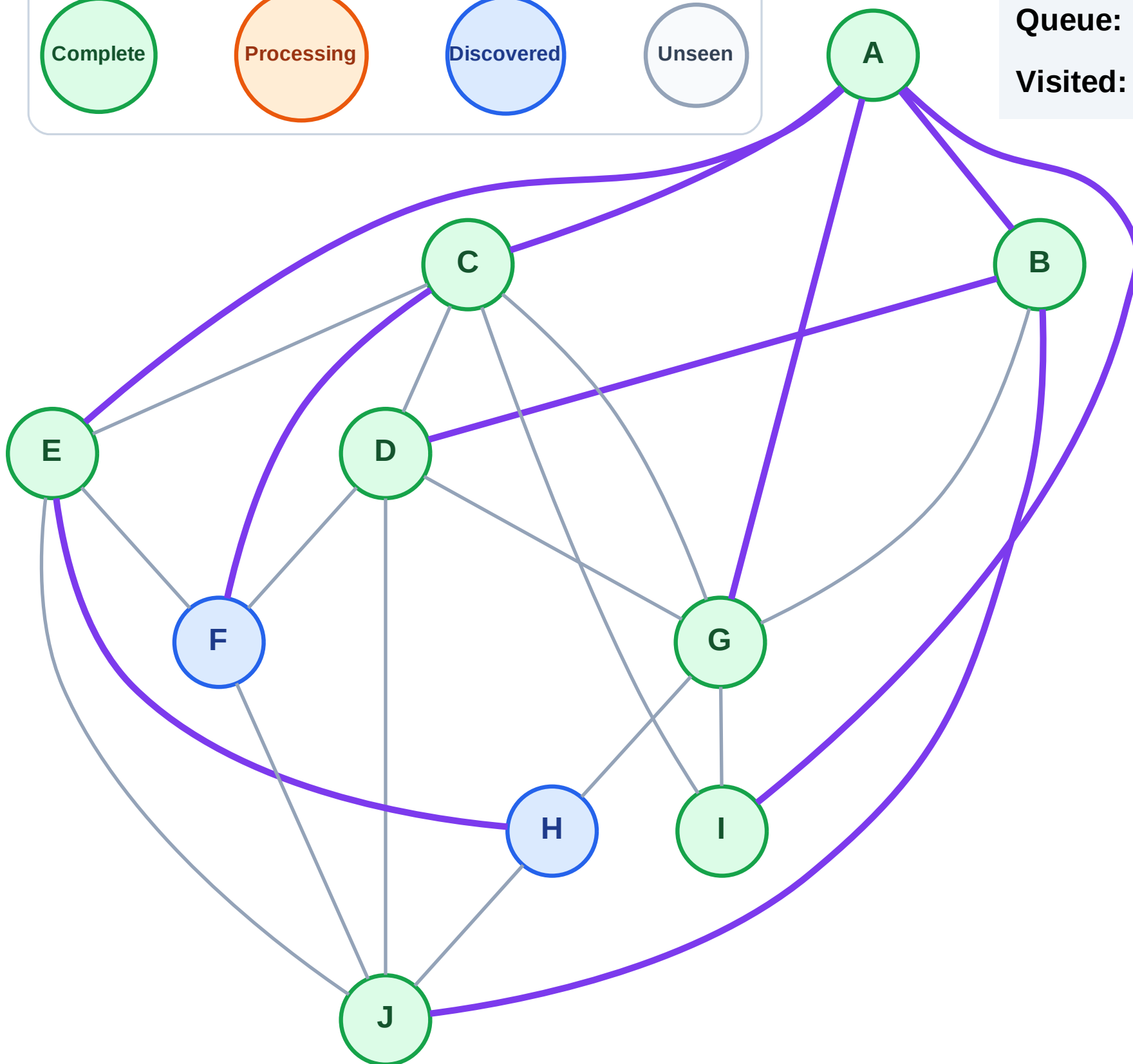
Step 17: No new neighbors from J

Legend



Queue: F → H

Visited: A, B, C, D, E, G, I, J



BFS Traversal

Step 18: Process node F

Legend

Complete

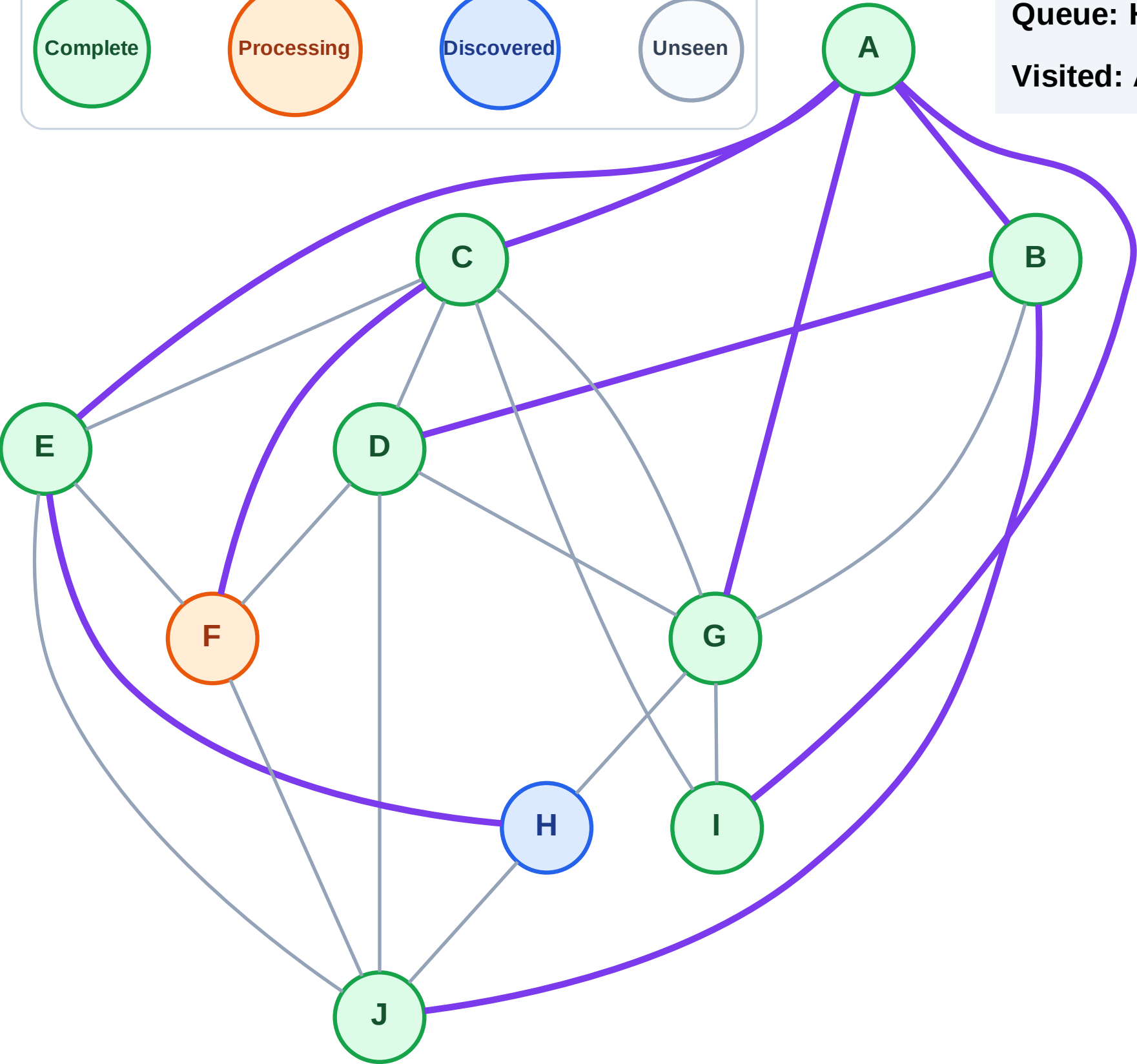
Processing

Discovered

Unseen

Queue: H

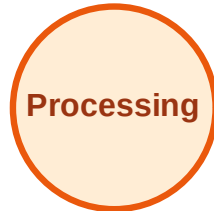
Visited: A, B, C, D, E, G, I, J



BFS Traversal

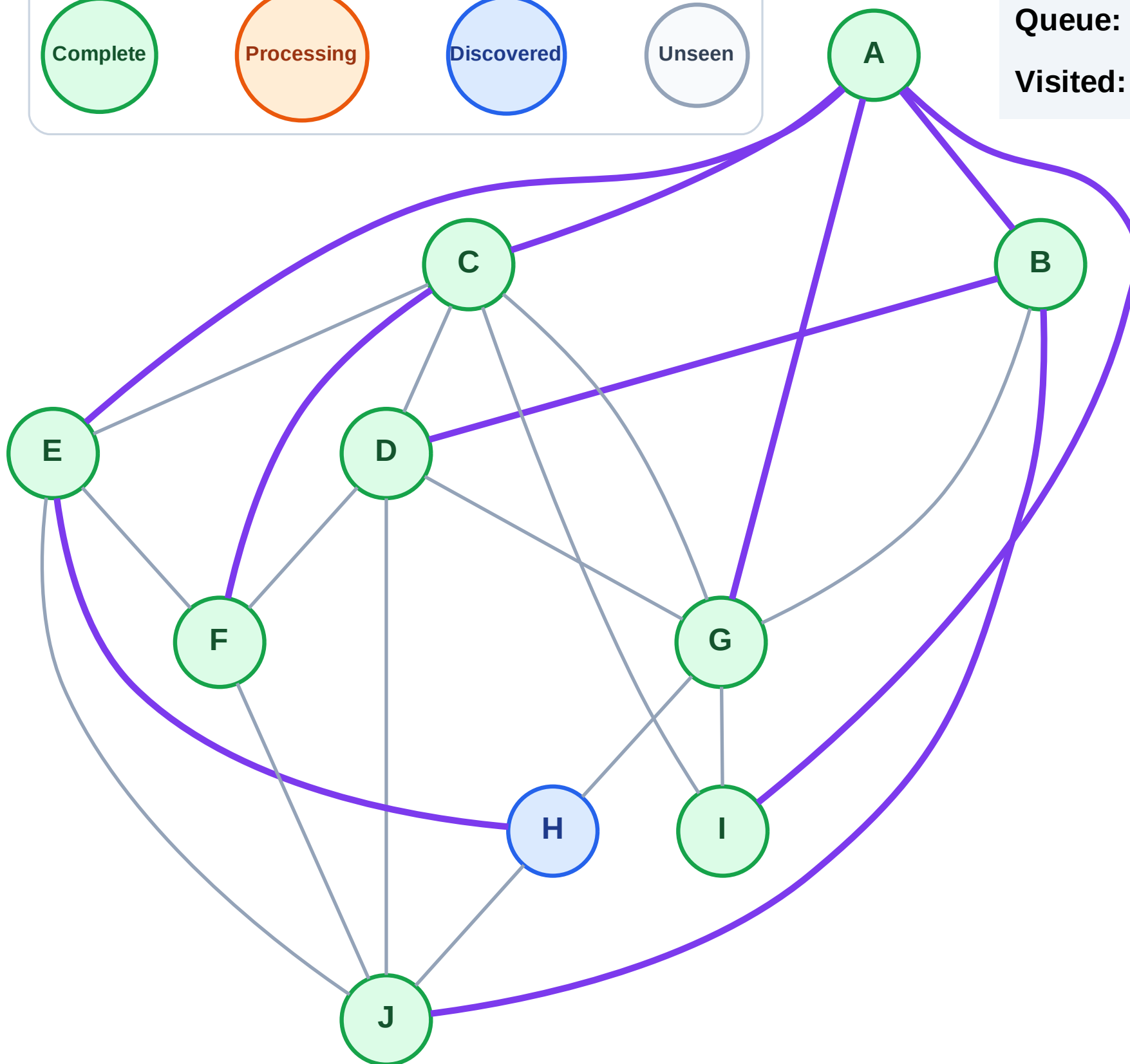
Step 19: No new neighbors from F

Legend



Queue: H

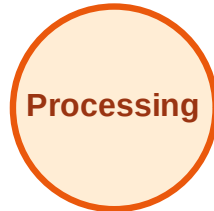
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

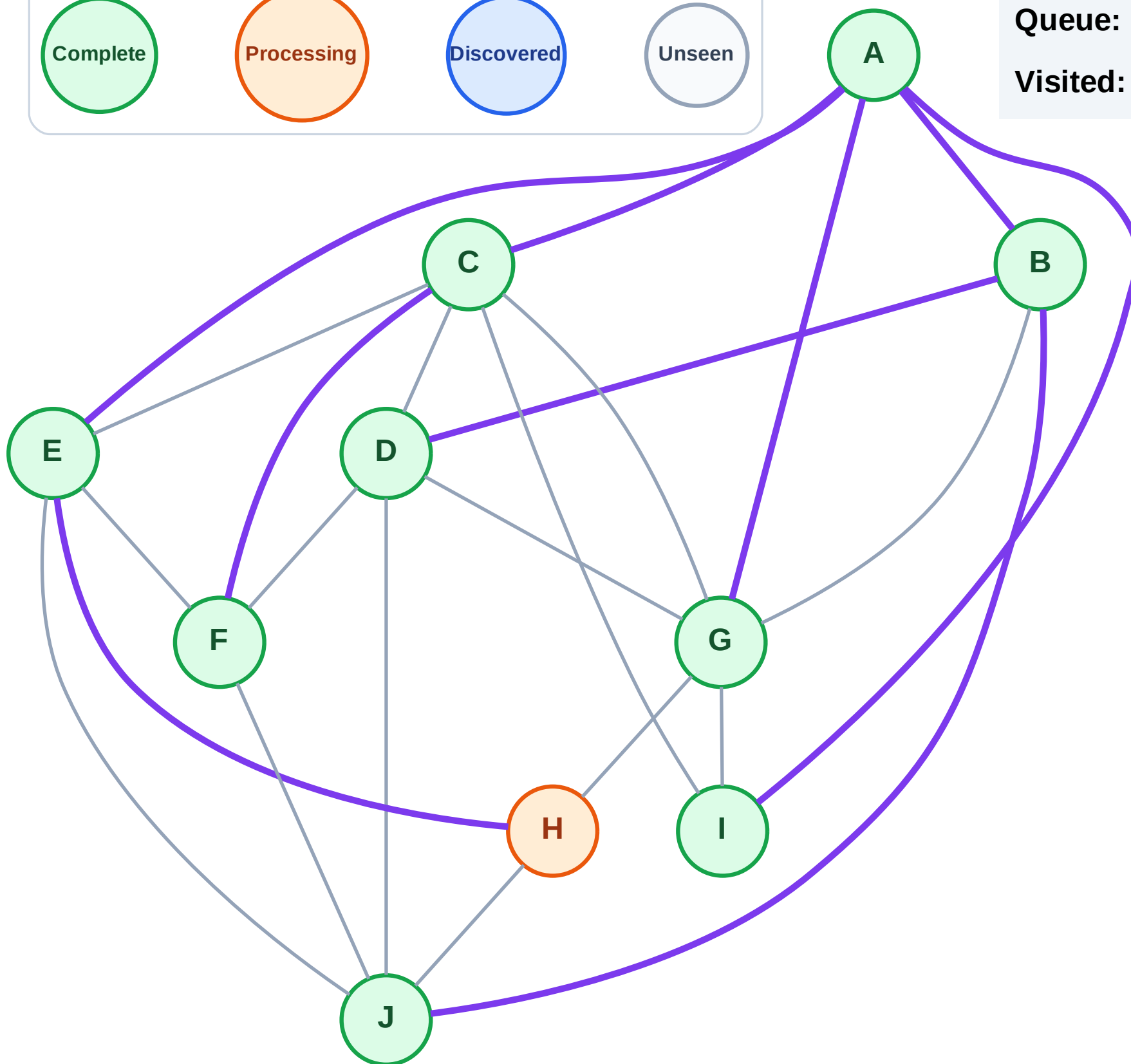
Step 20: Process node H

Legend



Queue: (empty)

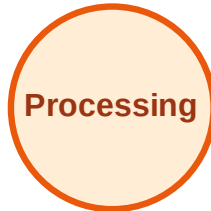
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

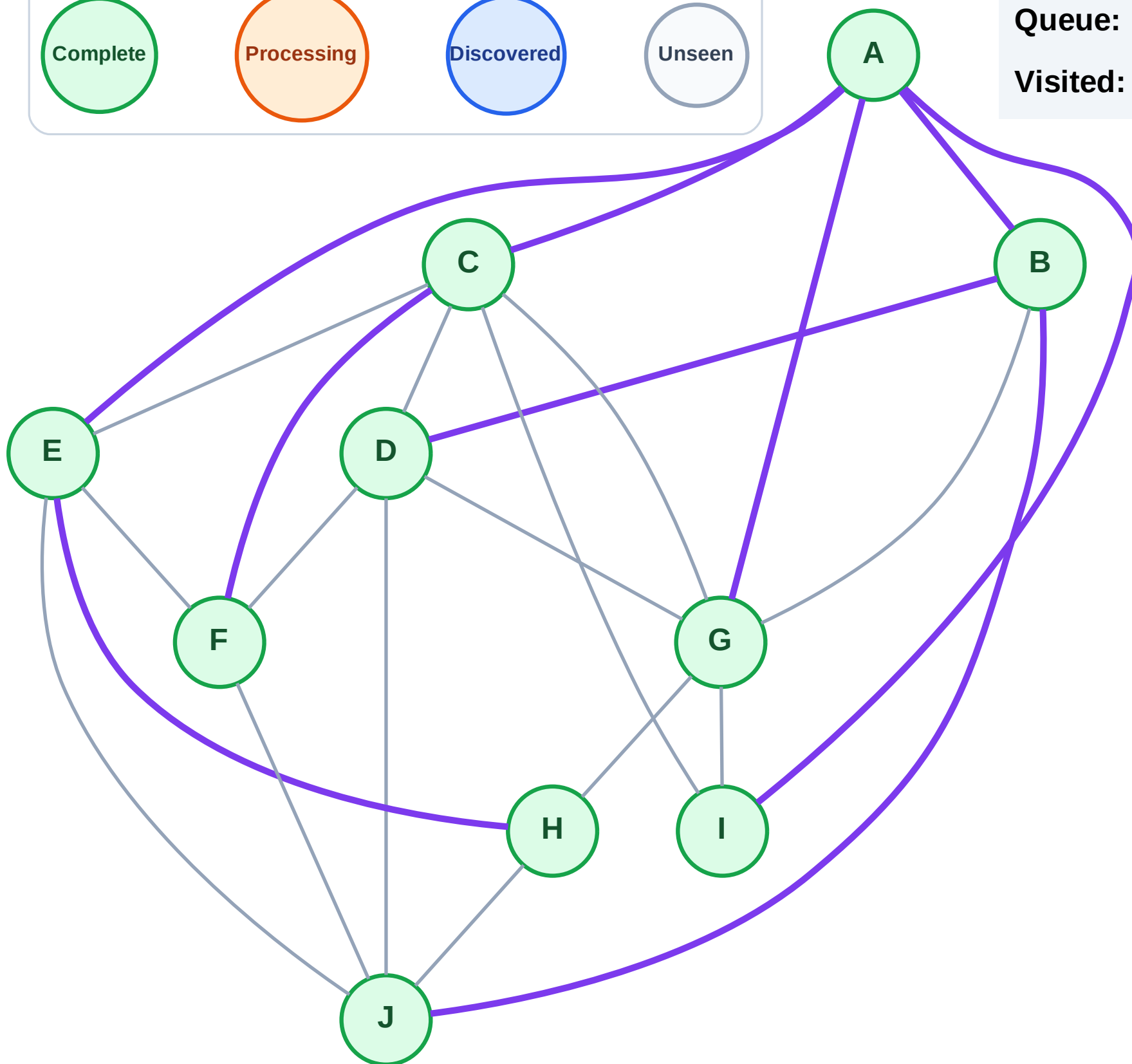
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

Complete

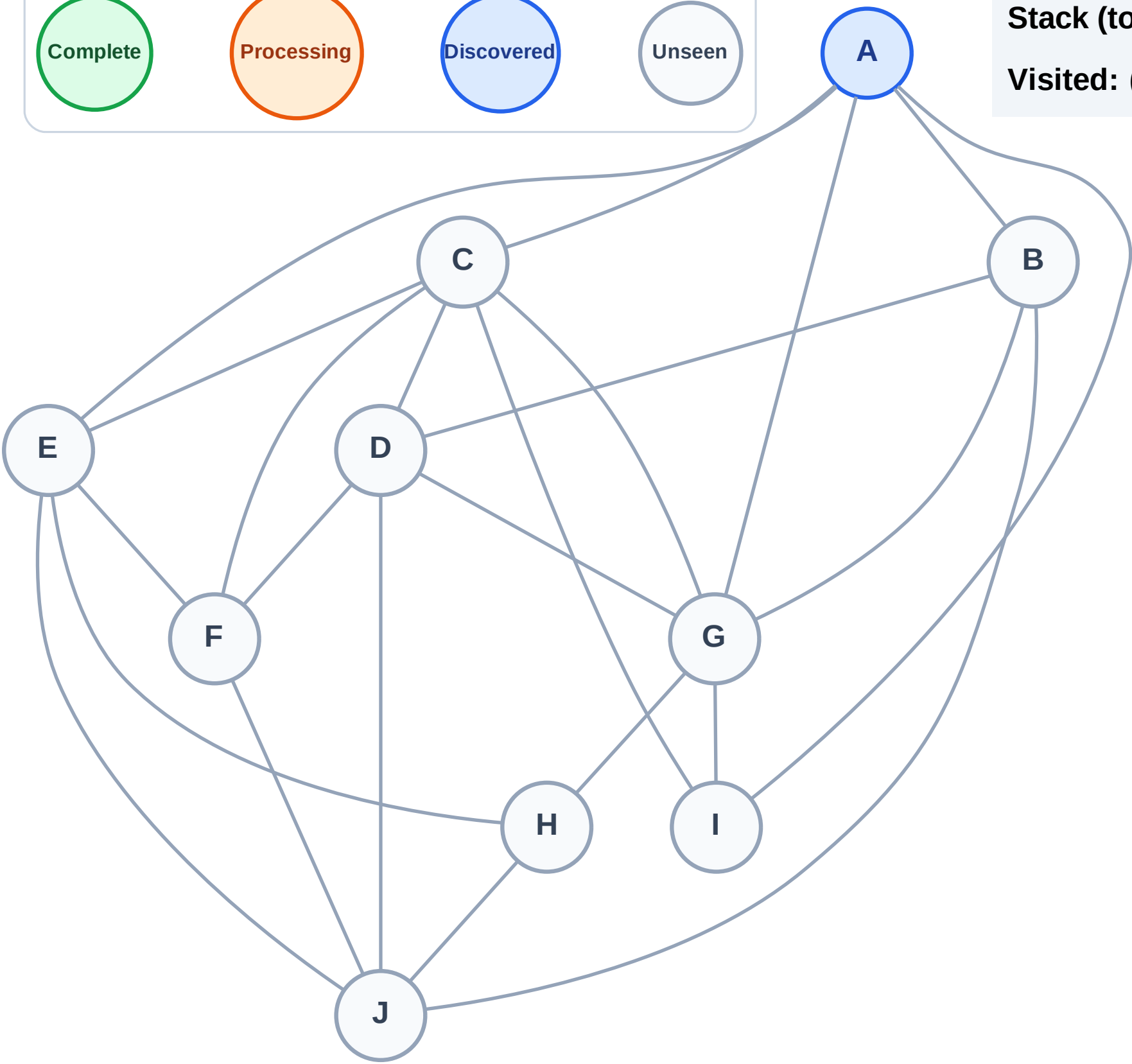
Processing

Discovered

Unseen

Stack (top at right): A

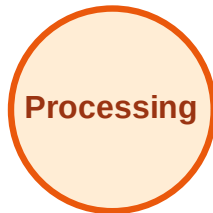
Visited: (none)



DFS Traversal

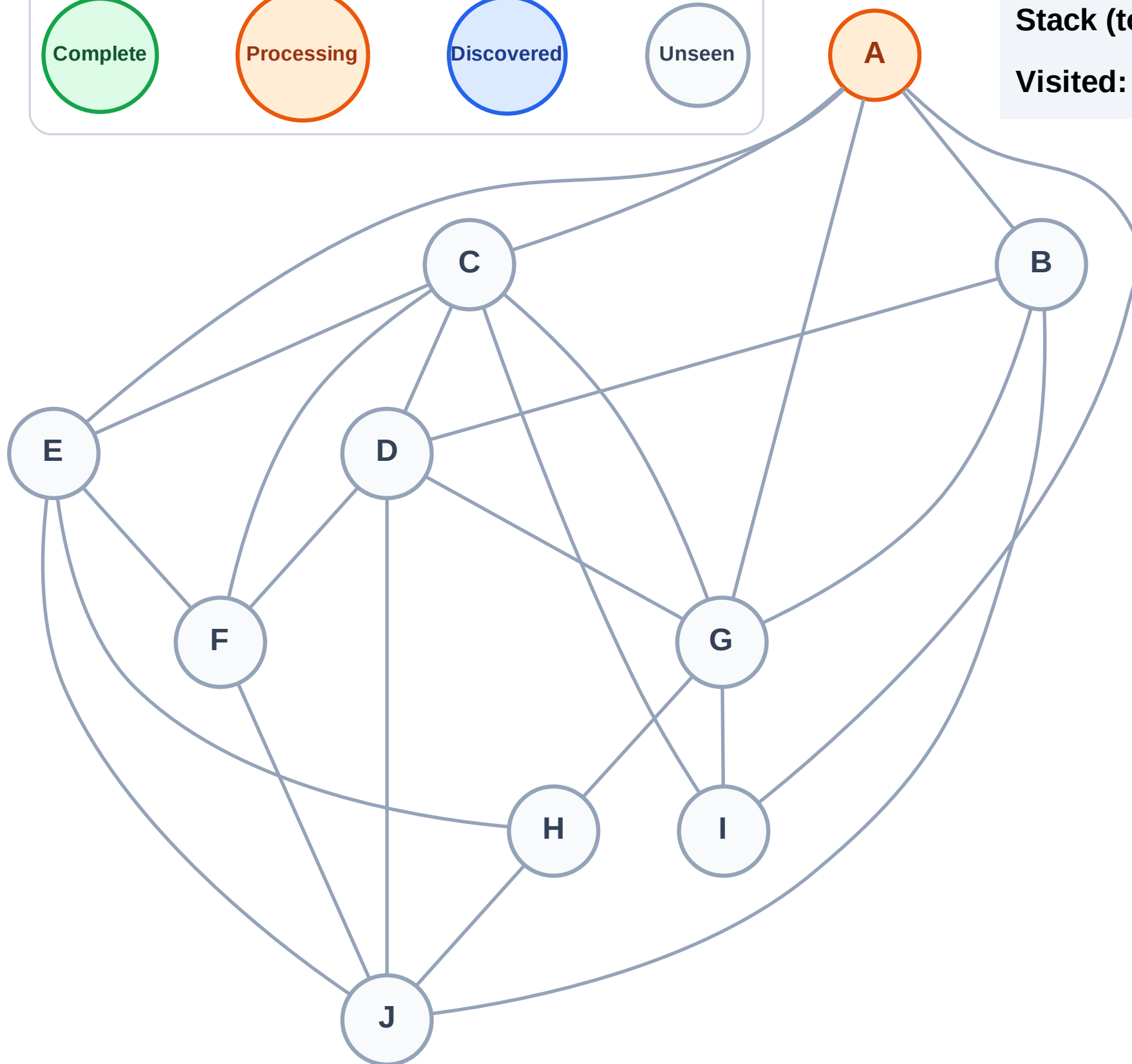
Step 2: Process node A

Legend



Stack (top at right): (empty)

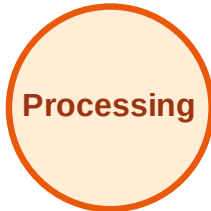
Visited: (none)



DFS Traversal

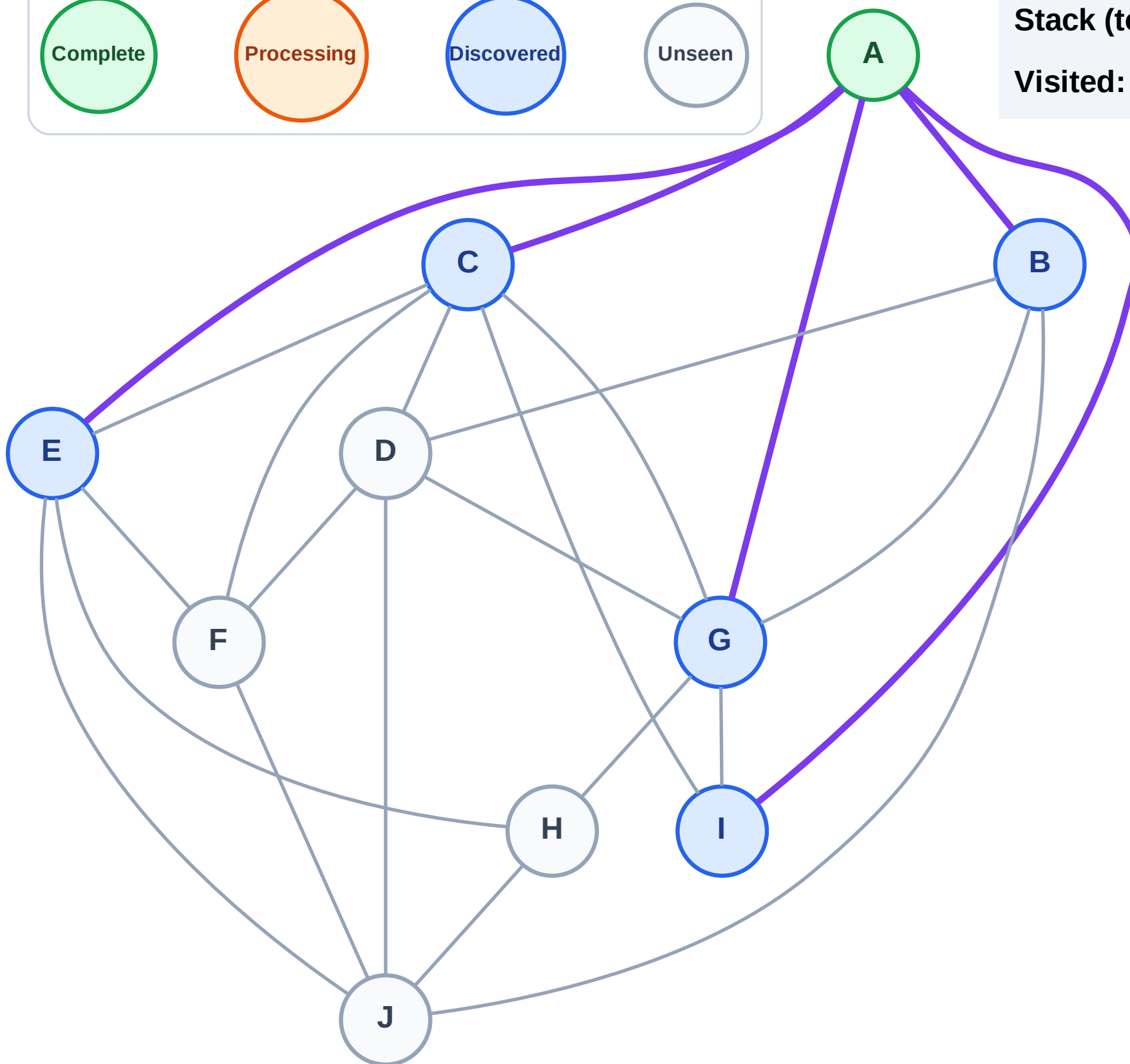
Step 3: Discover I, G, E, C, B from A

Legend



Stack (top at right): I | G | E | C | B

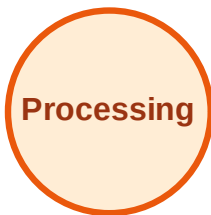
Visited: A



DFS Traversal

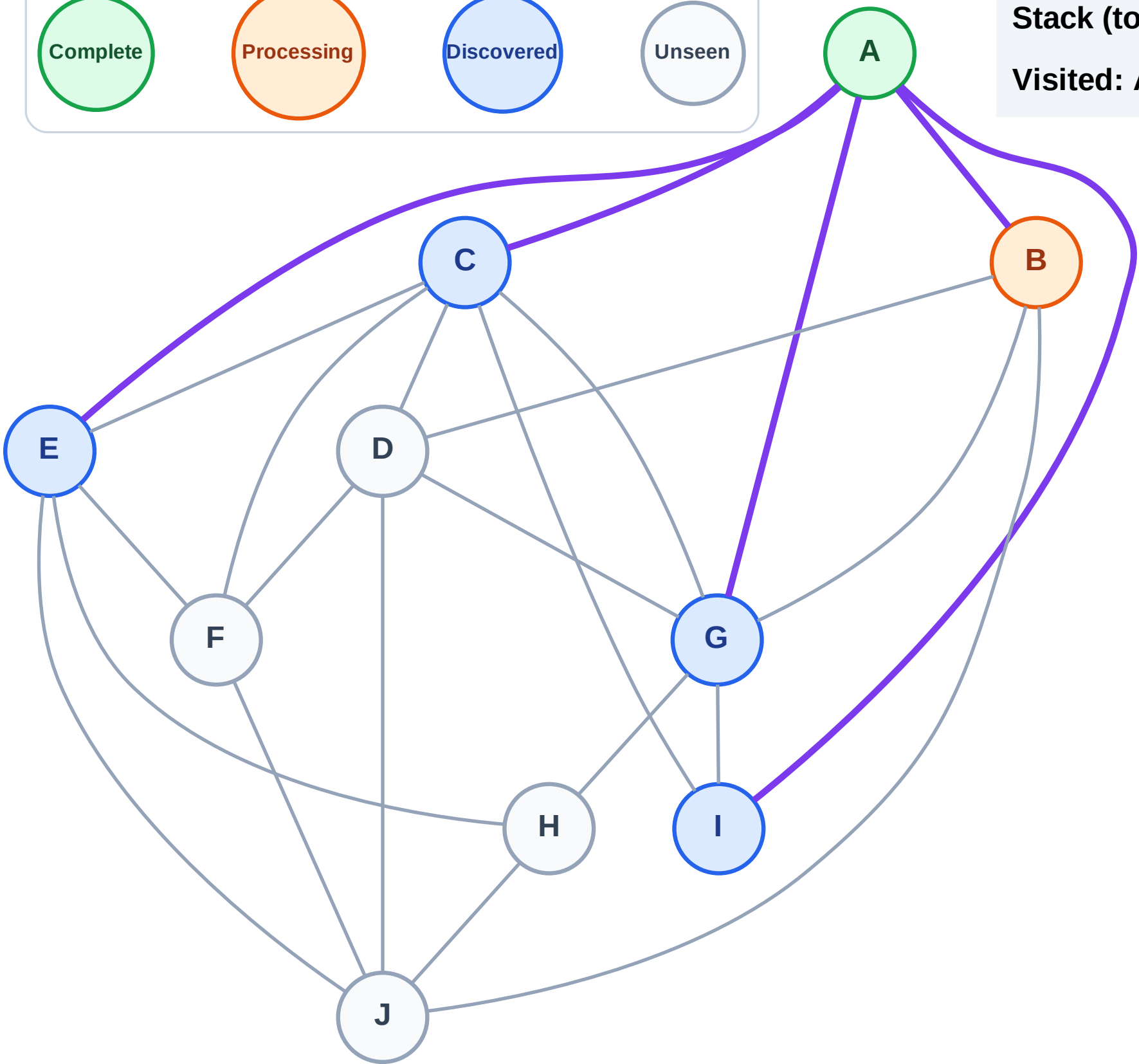
Step 4: Process node B

Legend



Stack (top at right): I | G | E | C

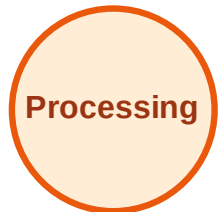
Visited: A



DFS Traversal

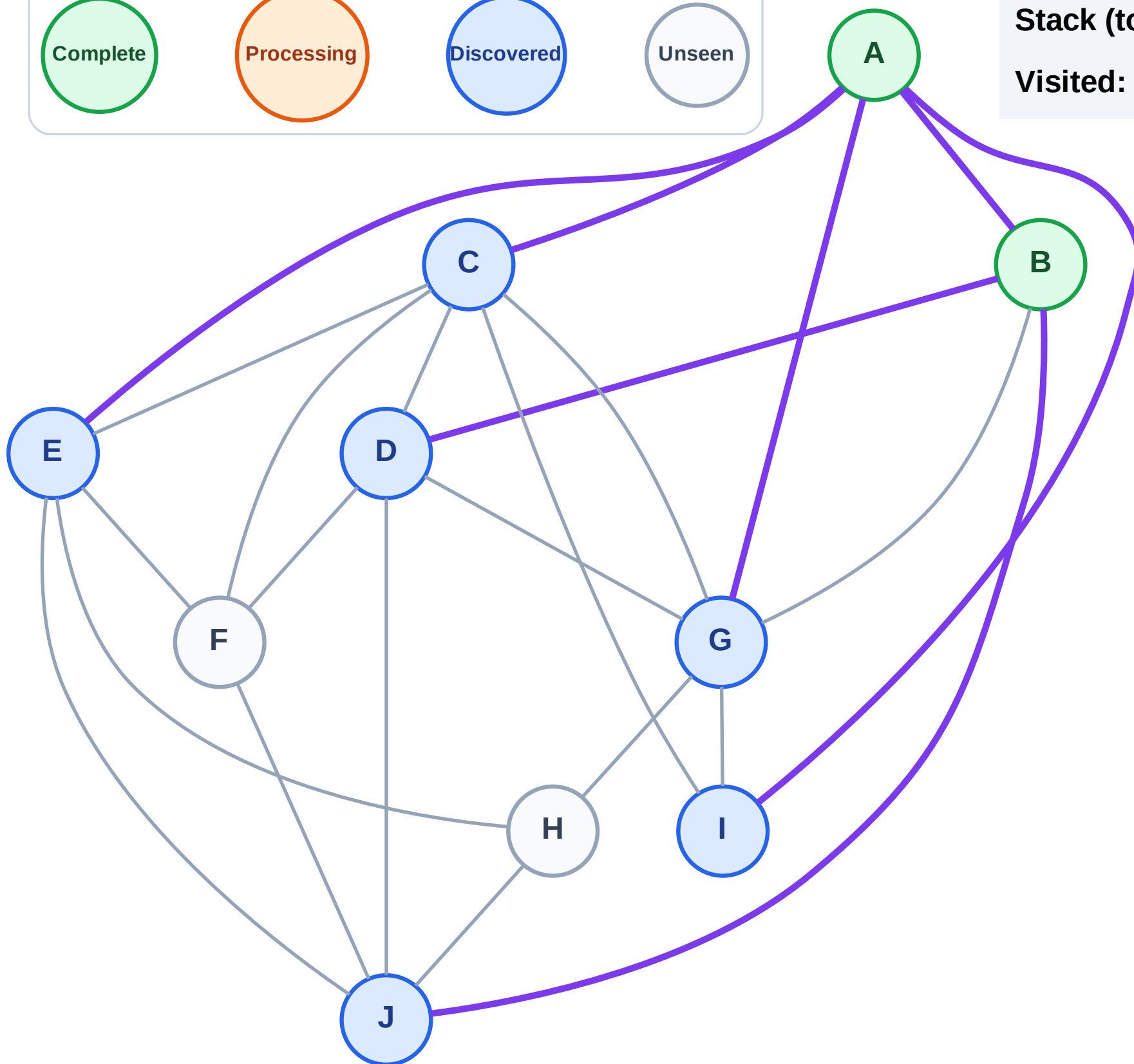
Step 5: Discover J, D from B

Legend



Stack (top at right): I | G | E | C | J | D

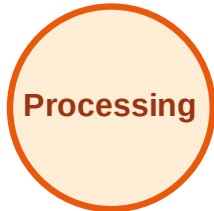
Visited: A, B



DFS Traversal

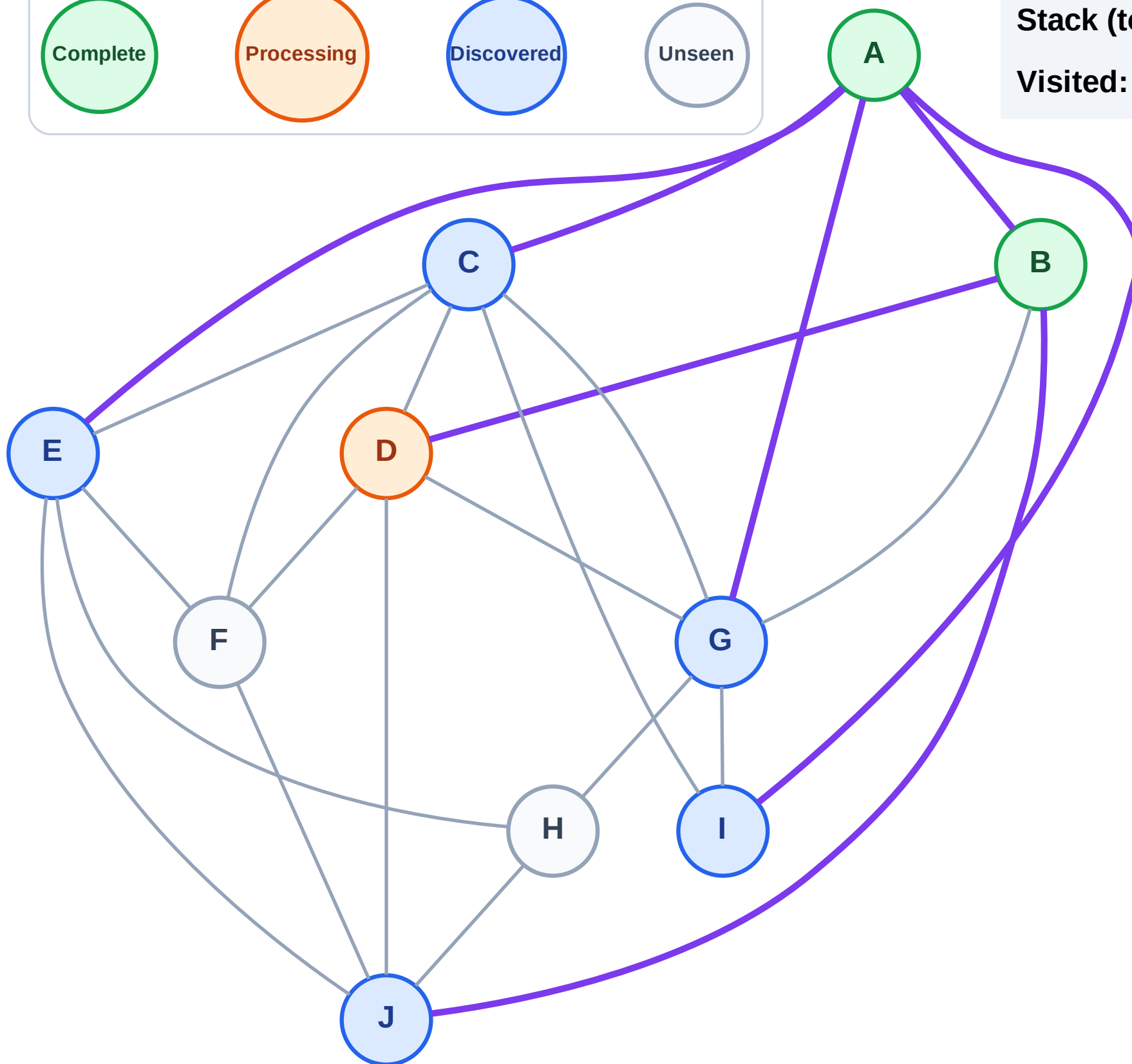
Step 6: Process node D

Legend



Stack (top at right): I | G | E | C | J

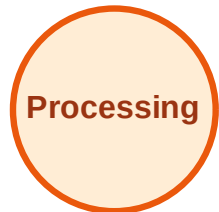
Visited: A, B



DFS Traversal

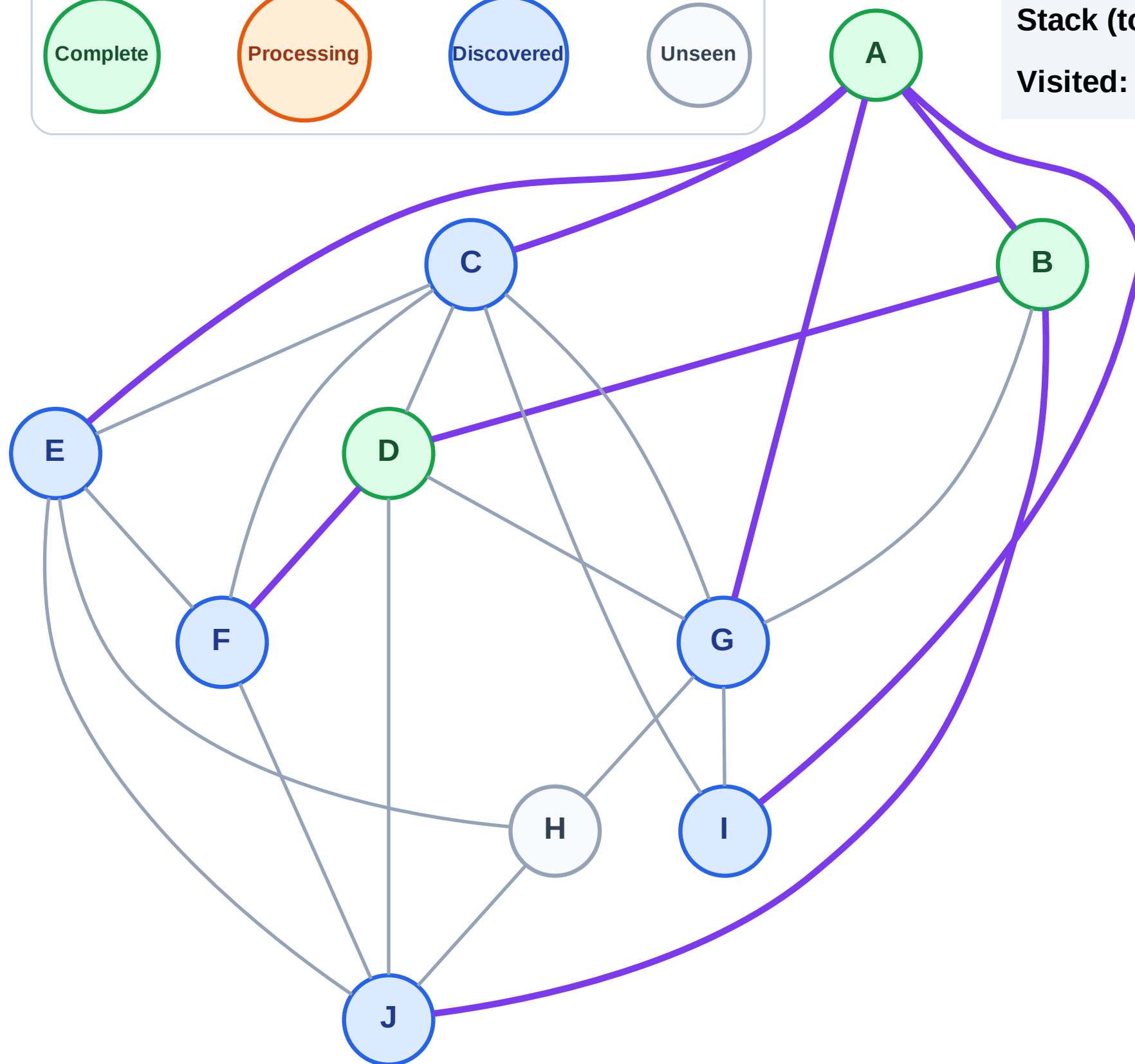
Step 7: Discover F from D

Legend



Stack (top at right): I | G | E | C | J | F

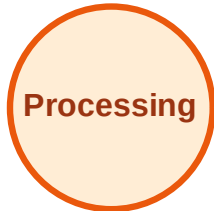
Visited: A, B, D



DFS Traversal

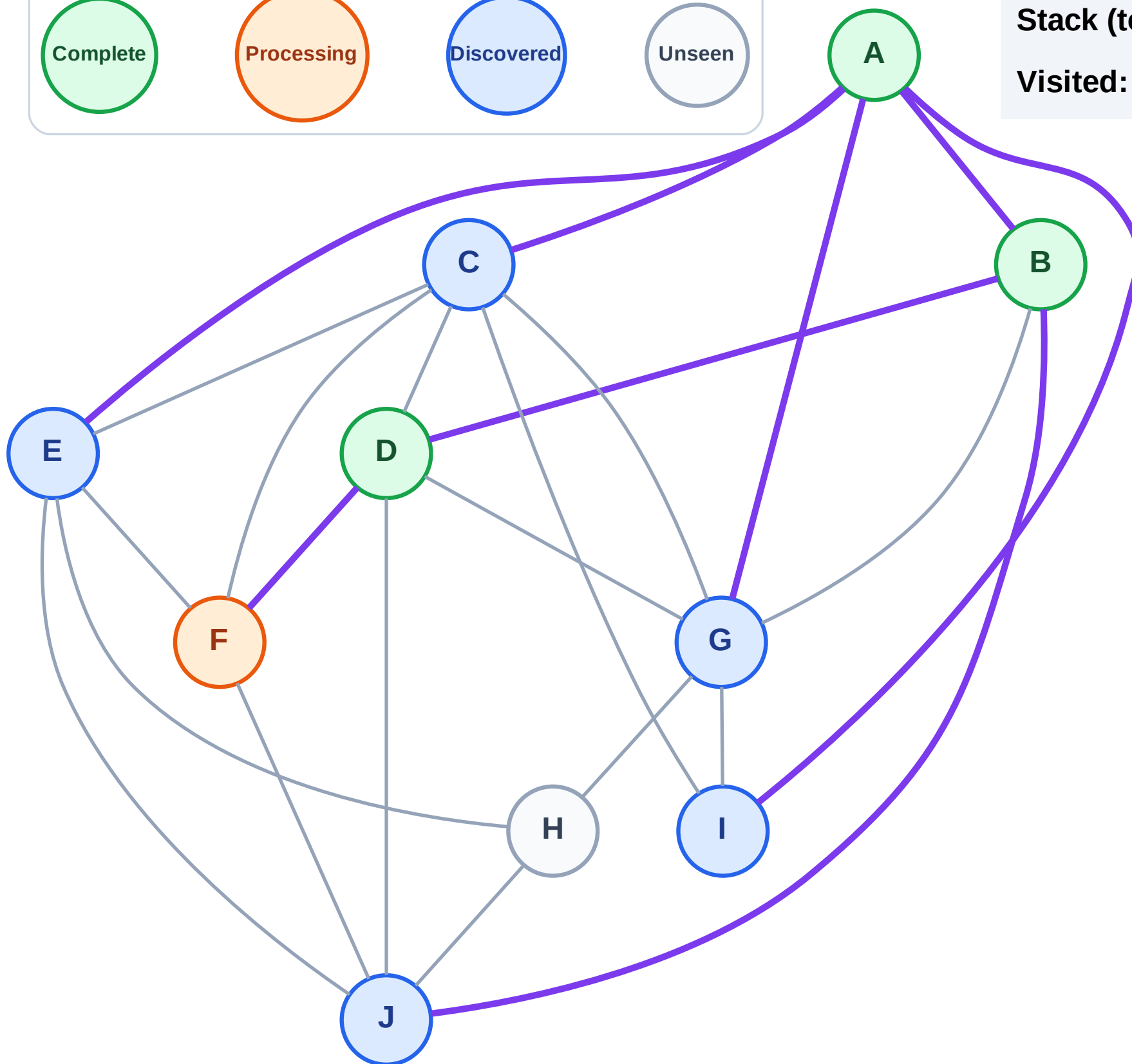
Step 8: Process node F

Legend



Stack (top at right): I | G | E | C | J

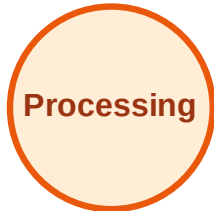
Visited: A, B, D



DFS Traversal

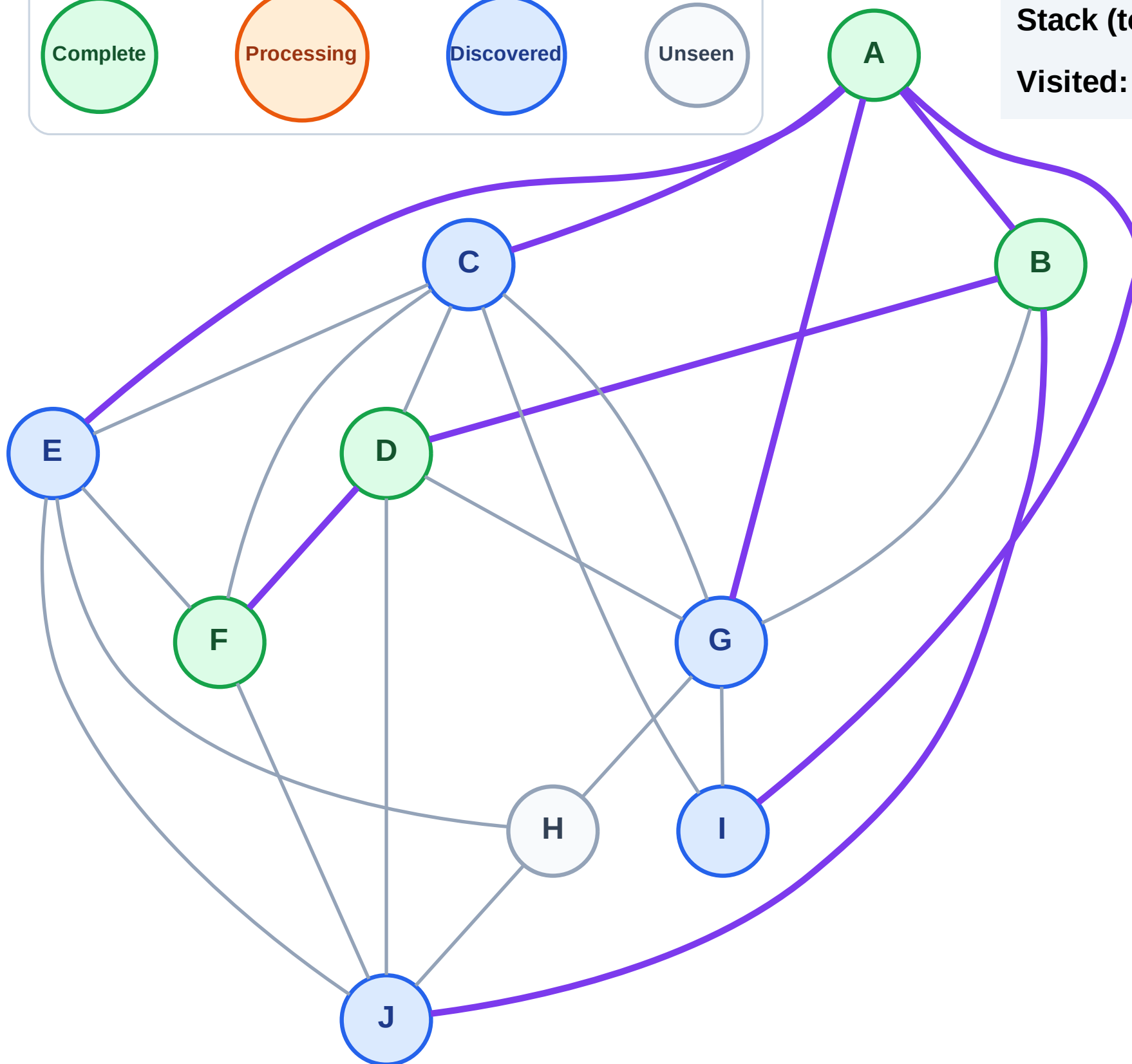
Step 9: No new neighbors from F

Legend



Stack (top at right): I | G | E | C | J

Visited: A, B, D, F



DFS Traversal

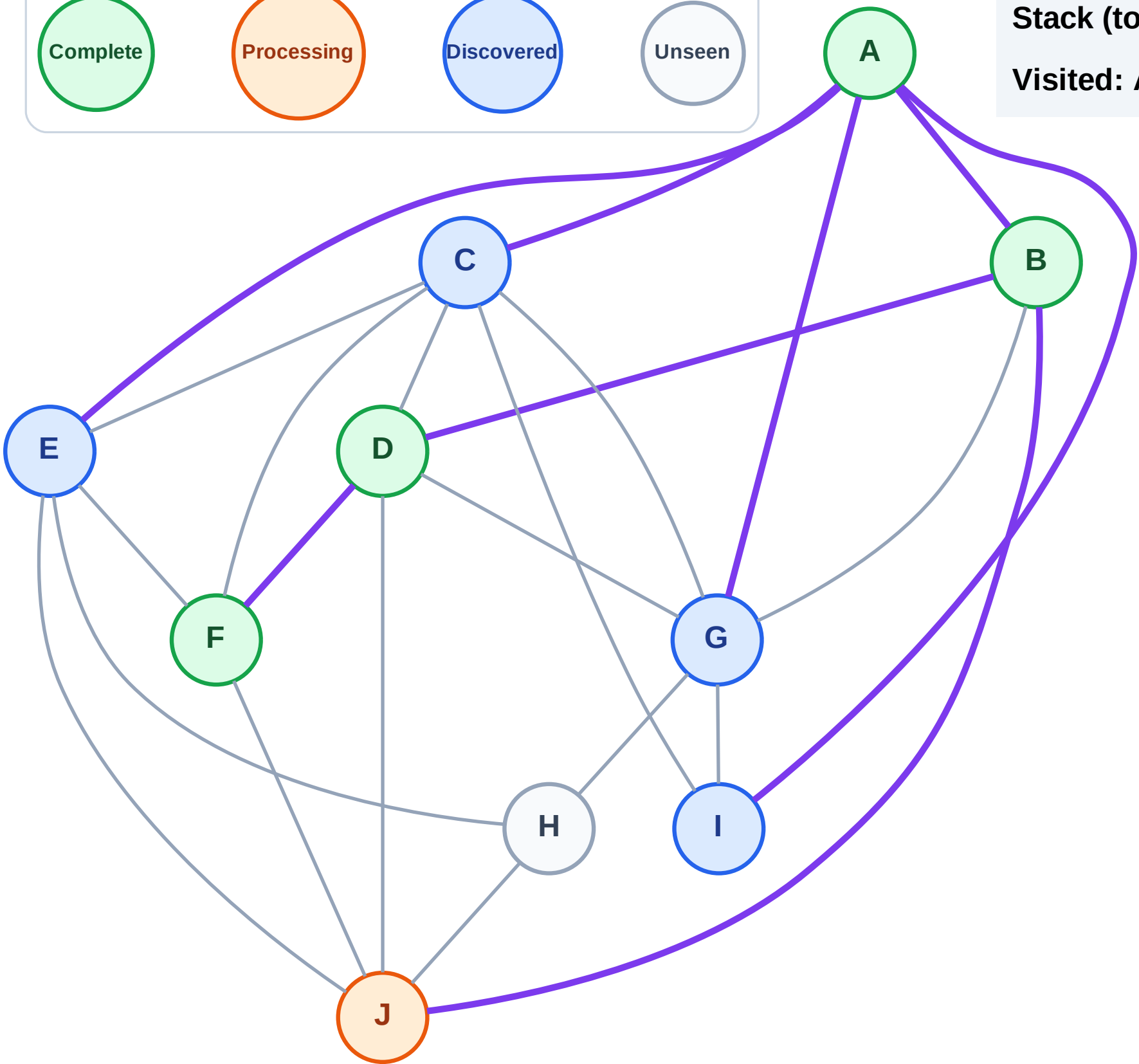
Step 10: Process node J

Legend



Stack (top at right): I | G | E | C

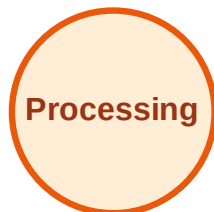
Visited: A, B, D, F



DFS Traversal

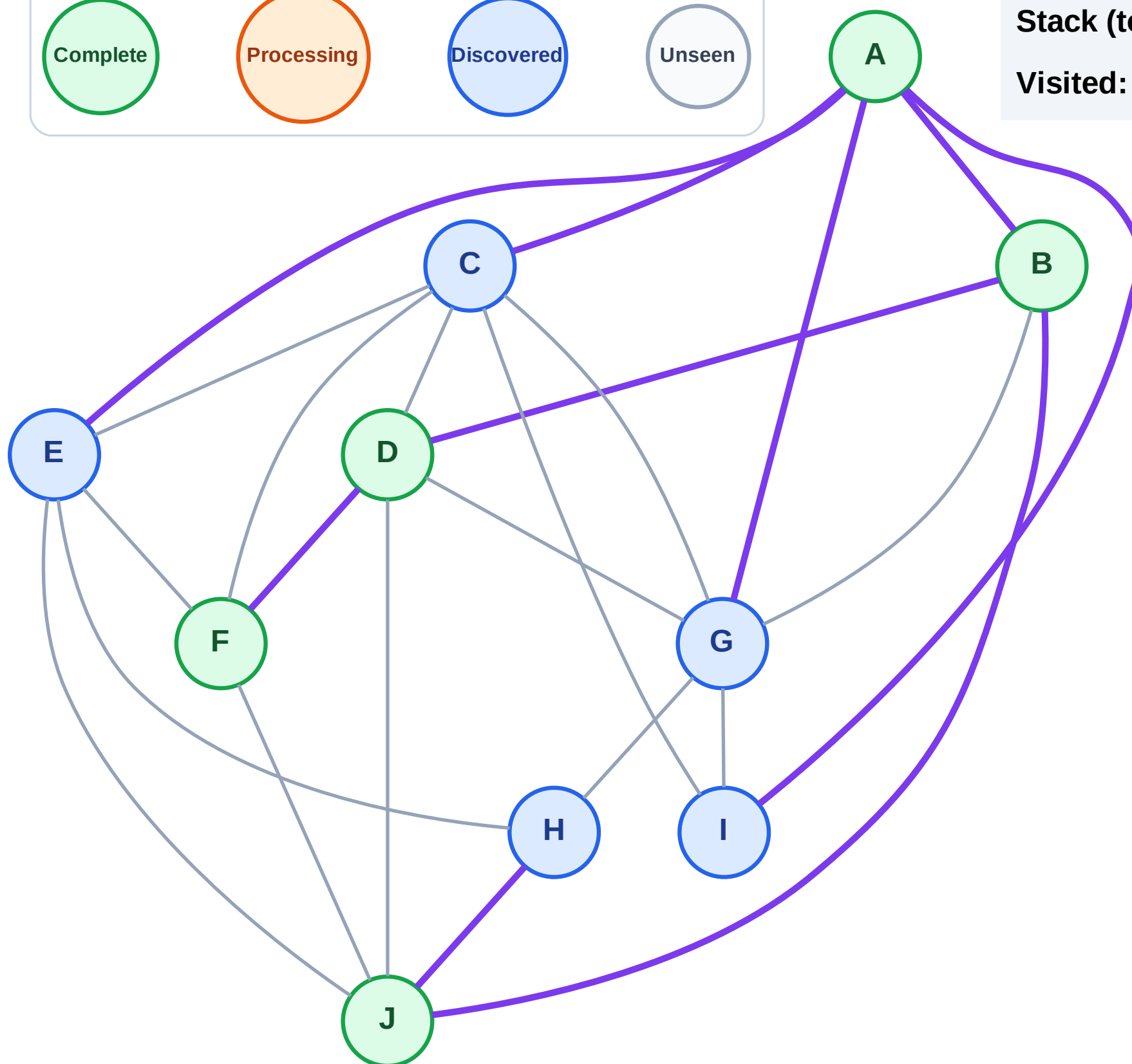
Step 11: Discover H from J

Legend



Stack (top at right): I | G | E | C | H

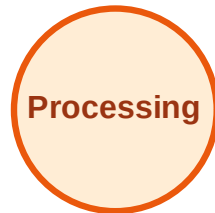
Visited: A, B, D, F, J



DFS Traversal

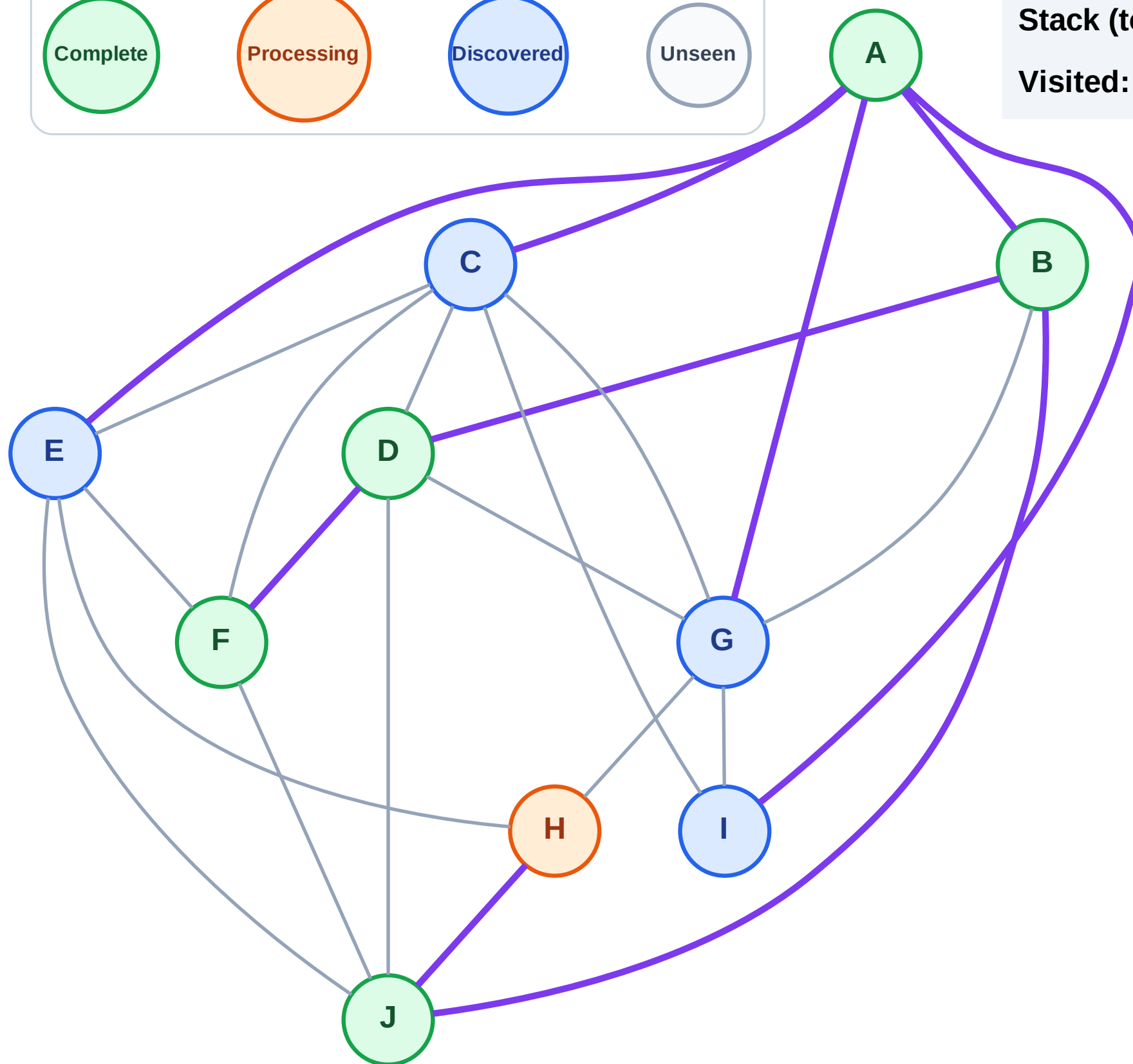
Step 12: Process node H

Legend



Stack (top at right): I | G | E | C

Visited: A, B, D, F, J



DFS Traversal

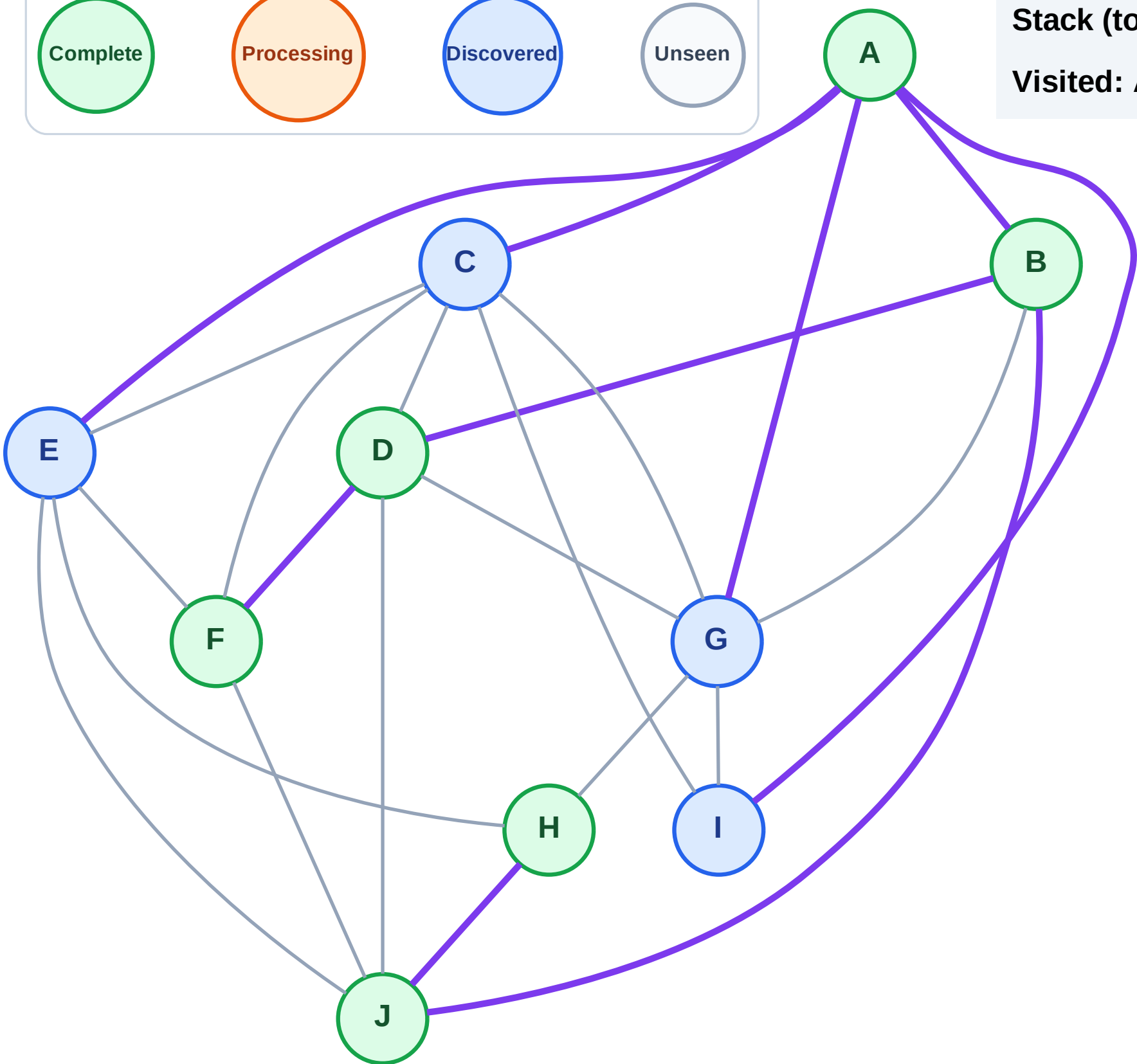
Step 13: No new neighbors from H

Legend



Stack (top at right): I | G | E | C

Visited: A, B, D, F, H, J



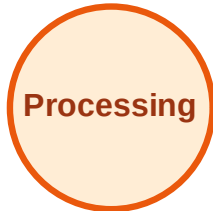
DFS Traversal

Step 14: Process node C

Legend



Complete



Processing



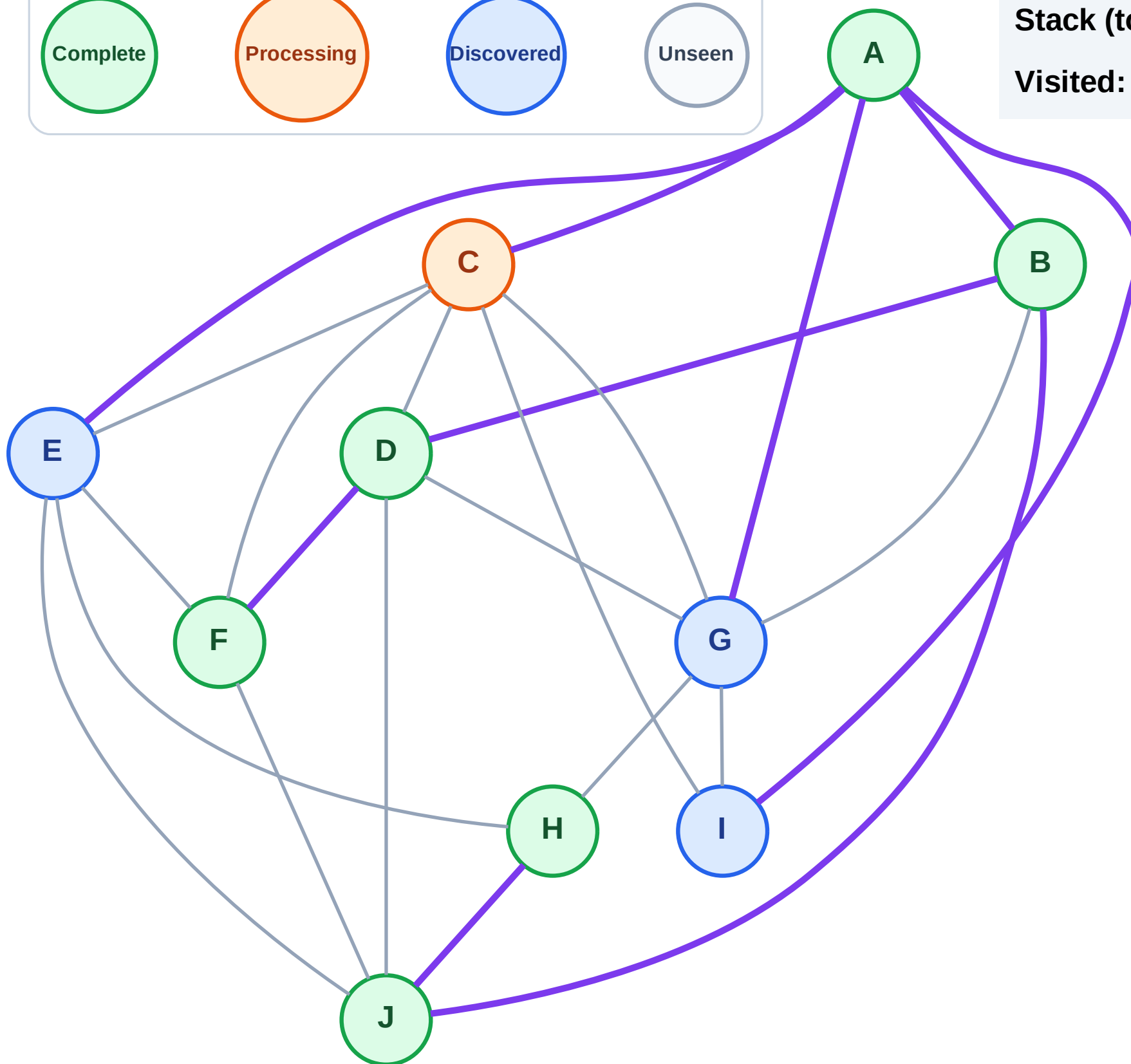
Discovered



Unseen

Stack (top at right): I | G | E

Visited: A, B, D, F, H, J



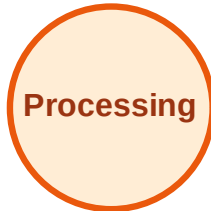
DFS Traversal

Step 15: No new neighbors from C

Legend



Complete



Processing



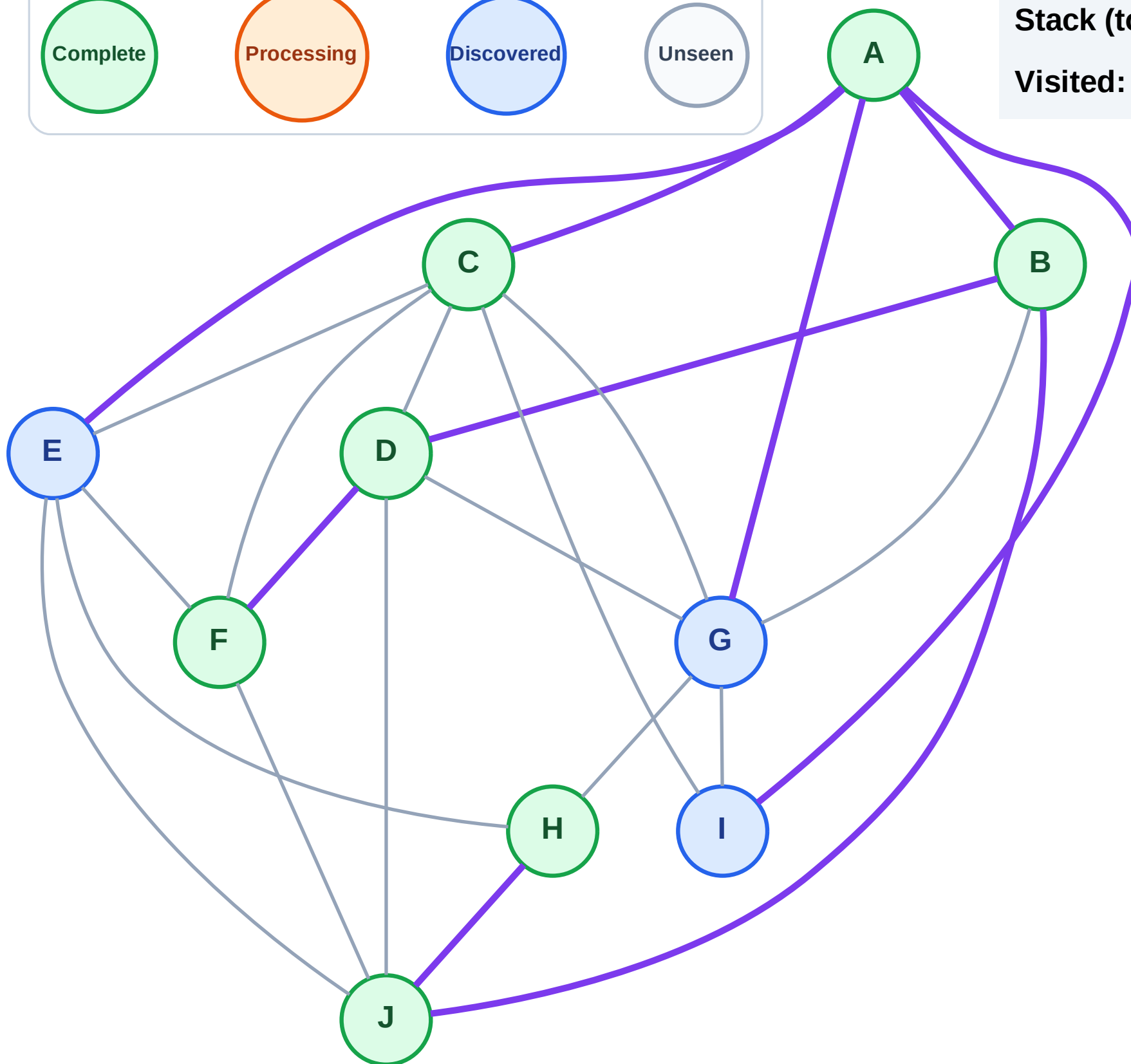
Discovered



Unseen

Stack (top at right): I | G | E

Visited: A, B, C, D, F, H, J



DFS Traversal

Step 16: Process node E

Legend

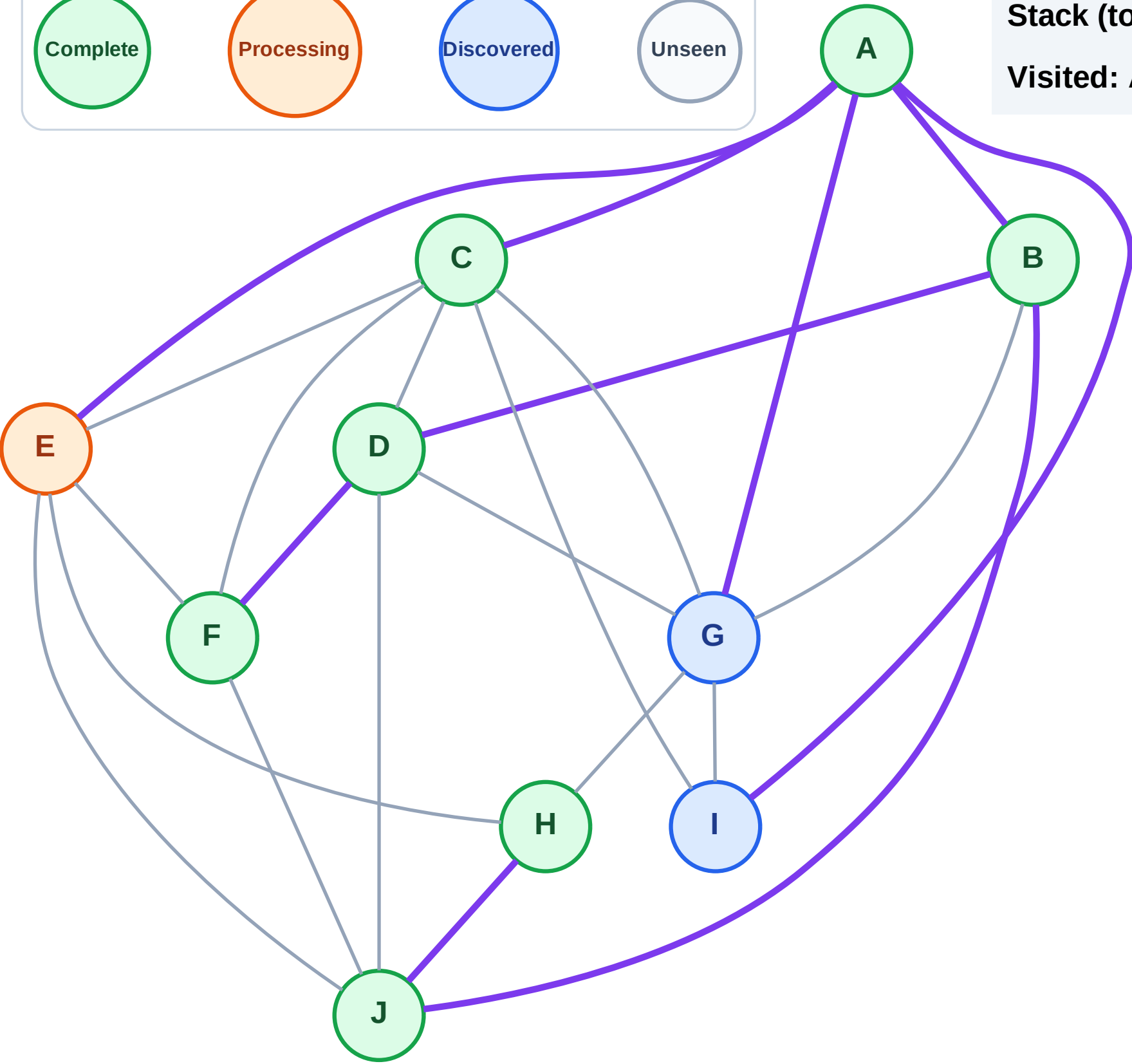
Complete

Processing

Discovered

Unseen

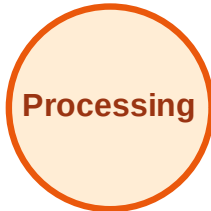
Stack (top at right): I | G
Visited: A, B, C, D, F, H, J



DFS Traversal

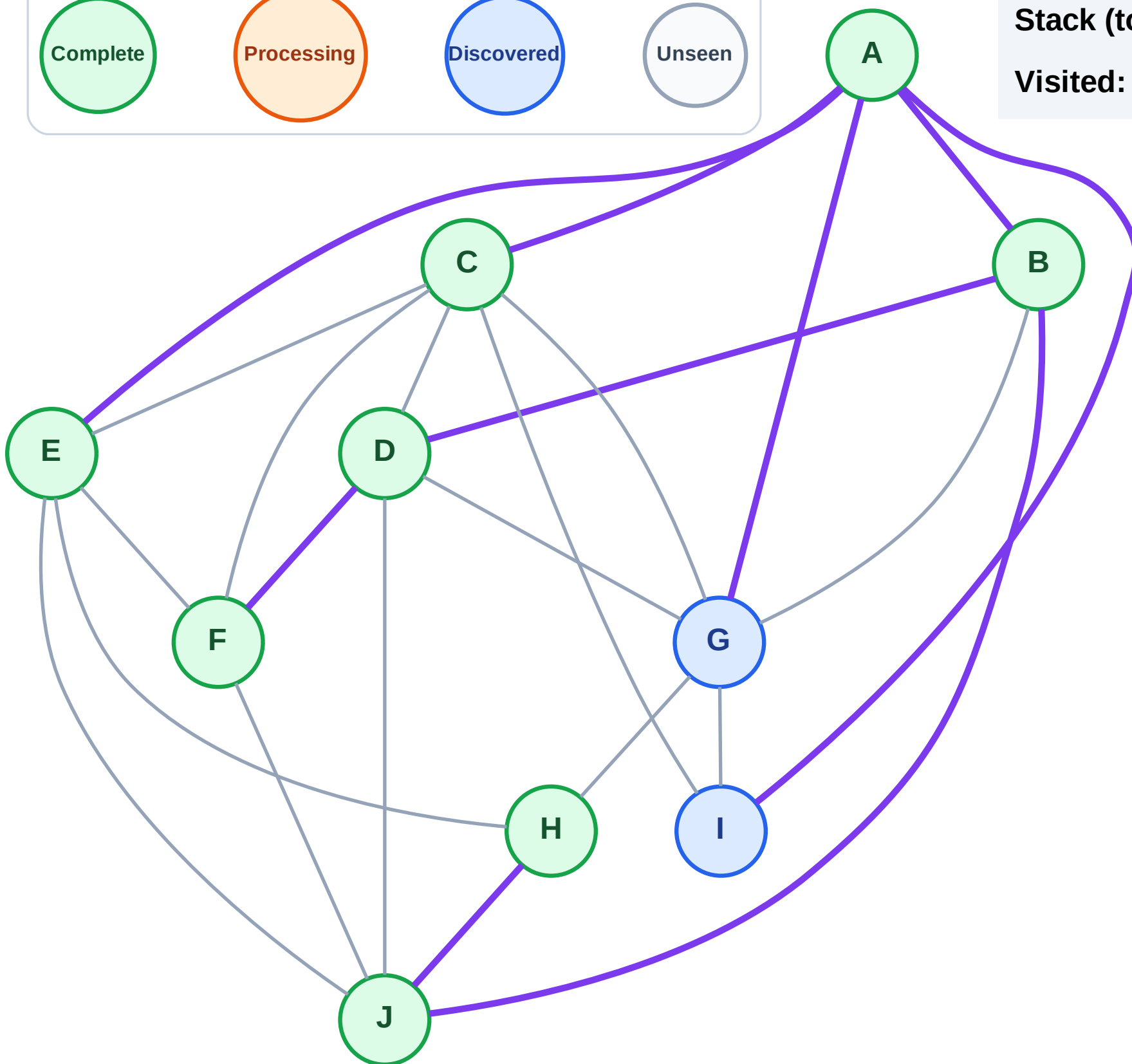
Step 17: No new neighbors from E

Legend



Stack (top at right): I | G

Visited: A, B, C, D, E, F, H, J



DFS Traversal

Step 18: Process node G

Legend

Complete

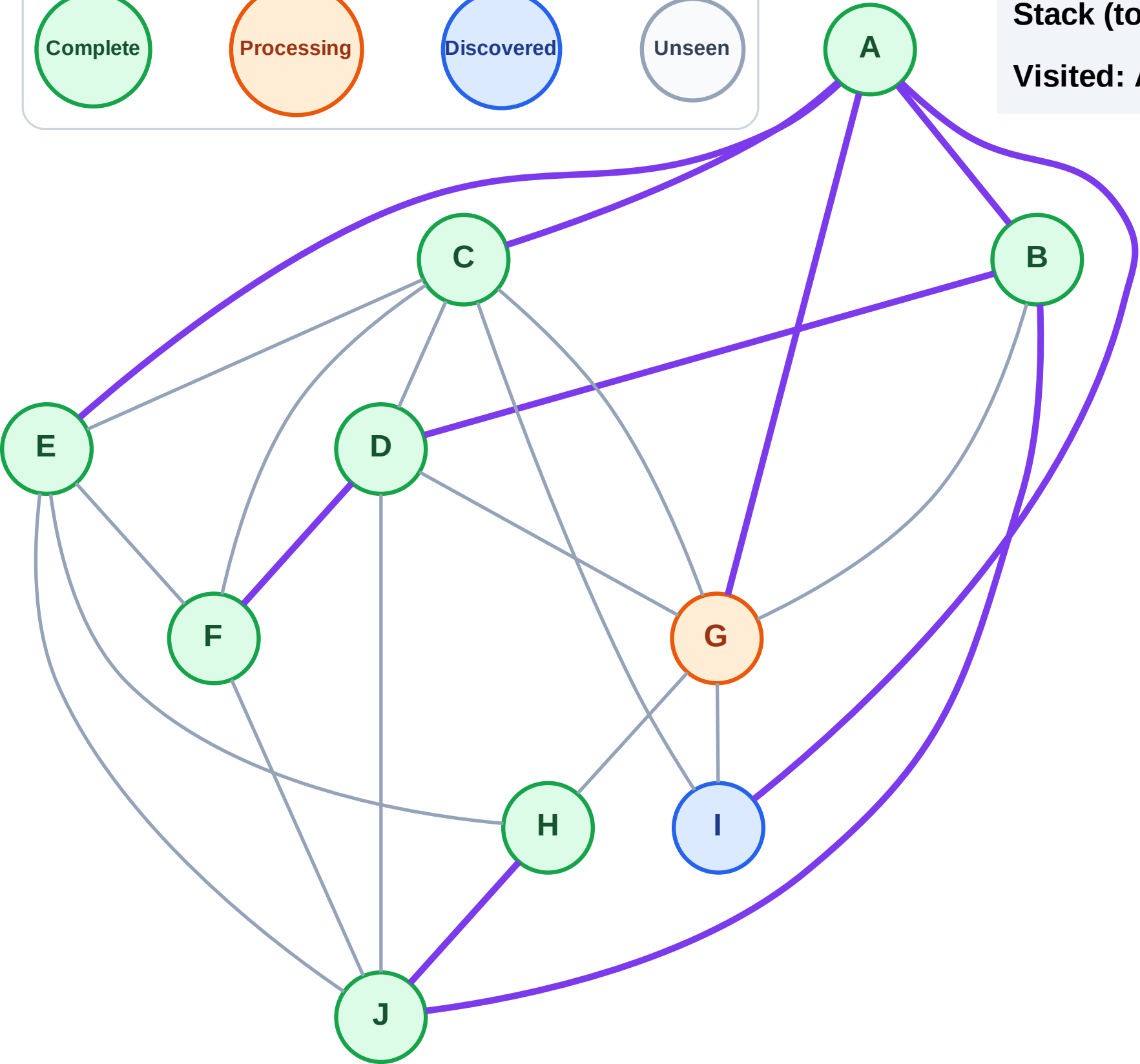
Processing

Discovered

Unseen

Stack (top at right): I

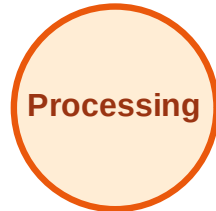
Visited: A, B, C, D, E, F, H, J



DFS Traversal

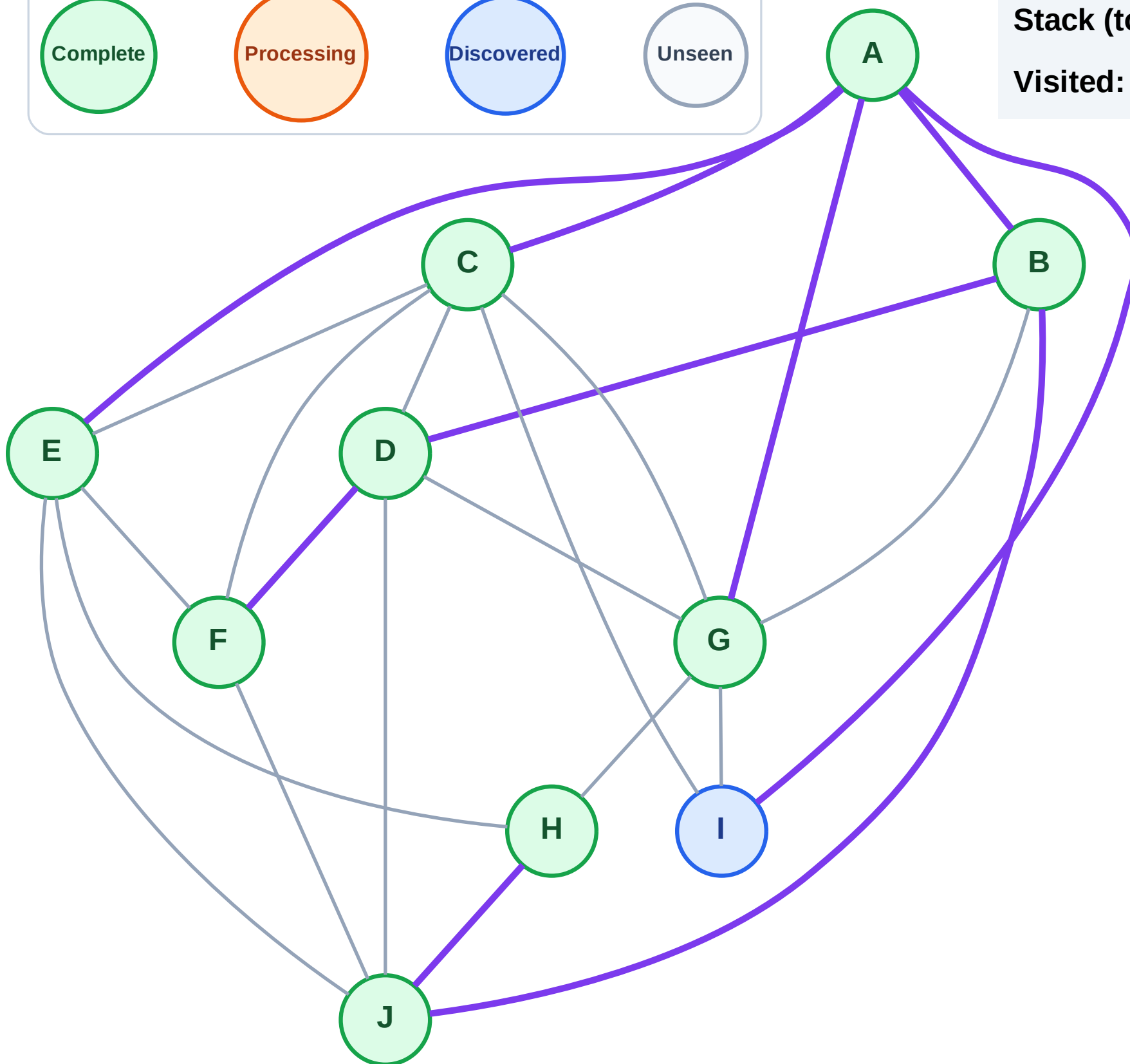
Step 19: No new neighbors from G

Legend



Stack (top at right): I

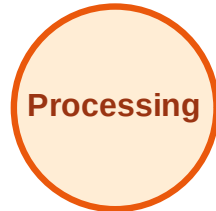
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

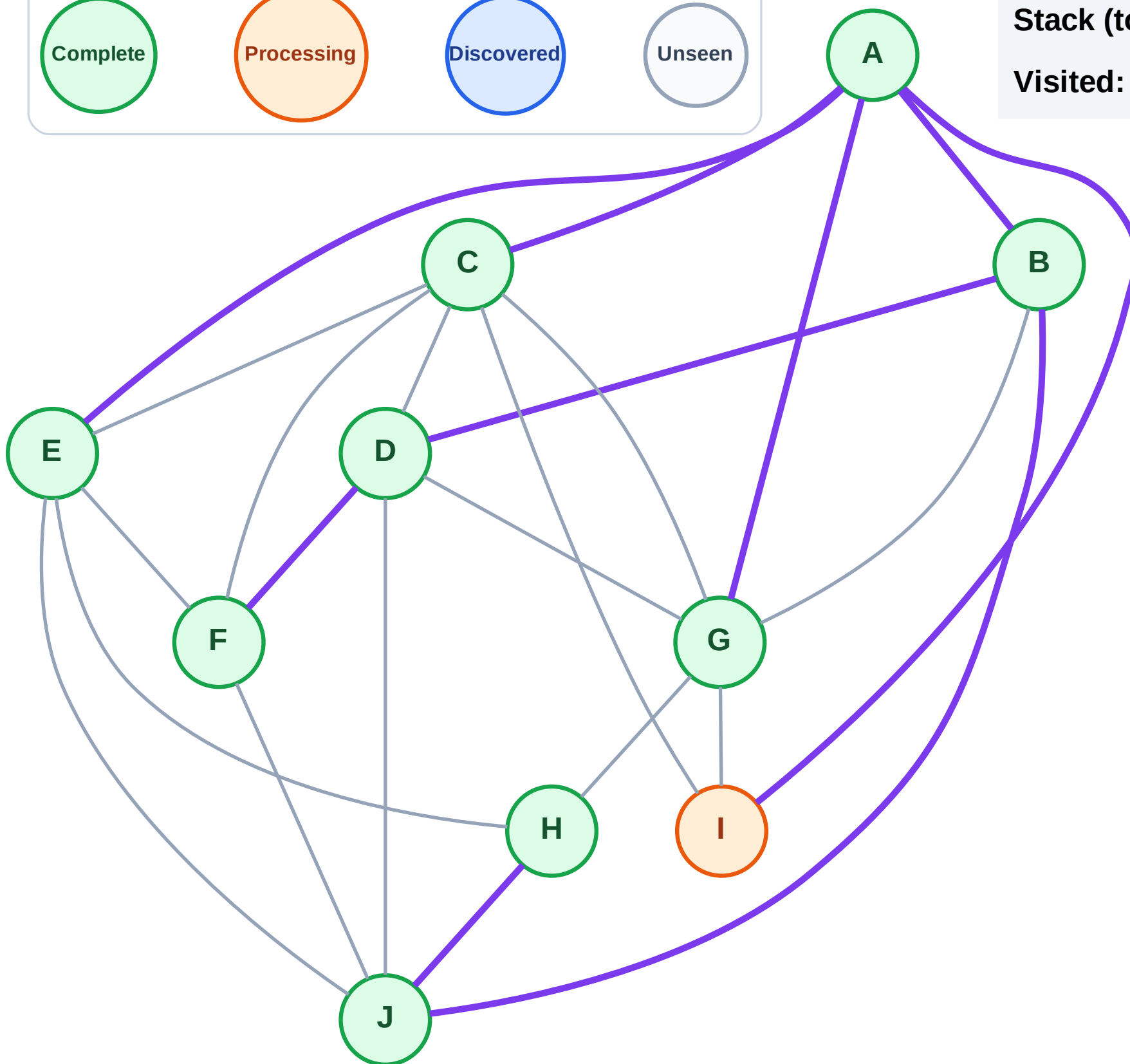
Step 20: Process node I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

